

Plasma Dynamics from the Unified 4D Model: A 4+1D Alternative to MHD with a Controlled MHD Limit and 4D Interaction Corrections

Trevor Norris

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Abstract

We present a 4+1D plasma interaction model derived from a unified bulk action with localized electromagnetism and an operational brane-observer map. The model provides exact bulk conservation laws, exact projected (brane) balance laws with explicit leakage terms, and controlled reductions that recover standard 3+1 Maxwell theory, two-fluid plasma equations, and the ideal-MHD induction equation. We then identify and organize the leading beyond-MHD corrections that arise when 4D interactions are retained: a dynamical scalar-photon sector A_w , brane–bulk charge/current exchange J^w , and finite-localization (KK/mode) corrections that modify propagation and topology transport. Finally, we outline a simulation architecture for massive plasma computations that evolves either the full 4+1D system or a mode-reduced Hermite basis in w , with diagnostics designed to quantify reconnection and topology change as genuine 4D transport rather than ad hoc resistivity.

1 Introduction and Motivation

Magnetohydrodynamics (MHD) is the dominant continuum framework for macroscopic plasma dynamics across laboratory, space, and astrophysical settings. In its standard regime of validity—quasi-neutrality, low-frequency dynamics compared to light and plasma frequencies, and length-scales large compared to kinetic scales—MHD provides a compact and remarkably predictive description. However, the most important (and often most computationally expensive) plasma phenomena frequently occur precisely where ideal MHD is structurally stressed: strong gradients, topological reorganization of field structure, and rapid energy conversion. The canonical example is magnetic reconnection, where ideal flux-freezing would prohibit topology change, yet observed plasmas change connectivity on dynamical timescales.

A large portion of “extended MHD” and kinetic plasma theory can be understood as principled ways to weaken ideal constraints by introducing additional degrees of freedom (Hall drift, electron pressure tensor, electron inertia, collisions, anomalous transport, etc.). In practice, many large-scale simulations still rely on closure choices whose physical meaning is indirect: reconnection is triggered by an imposed resistivity, by numerical diffusion, or by semi-empirical terms that are difficult to relate to a deeper ontology.

This paper develops an alternative starting point. Instead of taking 3D MHD as the fundamental description, we treat the observable 3D world as an *operational brane* that measures a projection of a conservative *bulk* dynamics living in four spatial dimensions (x, y, z, w) . The key claim is not that MHD is incorrect, but that it is a controlled *limit* of a more complete interaction model. In that parent theory, topological evolution that appears non-ideal from a brane perspective can arise from conservative transport into the extra spatial direction w and into transverse field structure (mode towers), rather than from phenomenological dissipation imposed by hand.

Concretely, we study a coupled 4+1D system consisting of: (i) a localized electromagnetic gauge field $A_M(t, \mathbf{x}, w)$ whose kinetic term is weighted by a localization function $Z(w)$, and (ii) a multi-species charged medium that may be represented either as a multi-field nonrelativistic matter sector (Schrödinger/GNLS-type) or as a two-fluid/kinetic plasma sector. The observable brane fields are defined by a separate projection kernel $W(w)$, emphasizing that “the brane” is not a boundary condition but a measurement map. This separation (bulk dynamics vs. brane observables) is essential: the bulk can remain conservative while the projected brane system becomes an open subsystem with explicit exchange terms.

1.1 What This Paper Claims (and What It Does Not)

Exact identities vs. controlled limits. We distinguish three categories of statements throughout:

1. **Exact identities** that follow from the bulk field equations and definitions (e.g. Bianchi identities, bulk charge conservation, exact projected balance laws with leakage sources).
2. **Controlled limits** in which the bulk system reduces to familiar 3+1D theories under explicit assumptions (e.g. zero-mode dominance in w , negligible transverse current J^w , quasi-neutrality, and small electron inertia).
3. **Regime statements** that specify when particular correction terms are expected to matter (e.g. when w -gradients are excited, when the scalar-photon sector A_w becomes dynamical, or when a finite number of transverse modes fails to capture observed dynamics).

Deriving MHD is a requirement, not the endpoint. A central goal of the paper is to show that standard MHD emerges as a *recoverable* description. We provide (i) a controlled reduction from localized 4+1D electromagnetism to effective 3+1D Maxwell theory on the brane, and (ii) a controlled reduction from a two-fluid brane plasma model to the ideal-MHD induction equation. These reductions are algebraic and are additionally verified by a symbolic reduction script (not included in the paper as source code).

The new physics is organized as explicit correction channels. The second central goal is to identify the leading beyond-MHD terms that the parent 4D model predicts when the controlled assumptions are relaxed. These corrections are not introduced phenomenologically; they arise from retained degrees of freedom and from the non-commutativity of projection with nonlinear dynamics. The paper organizes the most important extensions into:

1. a dynamical *scalar-photon* sector A_w and associated mixed field components $F_{\mu w}$, which drive bulk drift and new force channels;
2. explicit *brane-bulk exchange* via nonzero transverse current J^w , yielding open-system source terms in projected brane balance laws;
3. *finite-localization / transverse-mode* corrections (a KK/Hermite tower for Gaussian $Z(w)$) that provide computable short-range modifications and additional energy/topology storage channels;
4. a projected unresolved-stress contribution (a “topology stress tensor”) that quantifies how sub- w structure acts as an effective non-ideal term in the brane equations without requiring dissipation into heat.

Hybrid relativity (scope of validity). The electromagnetic sector is evolved as a relativistic hyperbolic system (wave propagation at speed c), while the baseline matter/plasma sector is nonrelativistic (as is standard in Schrödinger–Maxwell, two-fluid–Maxwell, and Vlasov–Maxwell simulation frameworks). The intent of this paper is to target MHD-like and extended-MHD-like regimes; fully relativistic plasma matter sectors (e.g. Dirac/Klein–Gordon descriptions or explicitly relativistic kinetic closures) are deferred to future work.

What we do not claim. We do not claim that the present paper completes a full kinetic microphysics program for all plasma regimes, nor that any single truncation of the transverse-mode tower is universally adequate. Instead, we provide a systematic parent model, a controlled reduction to known limits, and a hierarchy of correction terms with explicit simulation knobs that can be increased until convergence is achieved.

1.2 Roadmap

This paper is organized as follows. Section 2 defines the bulk/brane geometry, index conventions, and the operational projection map used to define brane observables. Section 3 presents the localized 4+1D electromagnetic sector, emphasizing the physical role of the A_w degree of freedom and the transverse-mode (KK/Hermite) structure induced by a localized kinetic weight $Z(w)$. Section 4 introduces the plasma matter sector (multi-species field, two-fluid, or kinetic) and the associated currents and conservation laws. Section 5 collects the coupled bulk system to be simulated and summarizes bulk invariants.

Section 6 presents the controlled reductions that recover effective 3+1 Maxwell theory on the brane and then recover the standard MHD limit from brane two-fluid dynamics. Section 7 is the main novelty section: it derives and organizes the beyond-MHD correction channels associated with A_w , J^w , finite localization (mode towers), and projected unresolved stresses. Section 9 describes how the model is intended to be simulated at scale, contrasting direct 4D-grid evolution with a mode-reduced 3D×modes scheme (Hermite in w) designed for massive computations. Section 10 outlines validation benchmarks in the controlled MHD limit and proposed tests that isolate genuinely 4D interaction effects. We conclude in Section 12 with a summary of the model hierarchy and near-term simulation targets.

2 Conventions, Geometry, and Operational Brane Observables

This section fixes notation and defines what is meant by “bulk” versus “brane” quantities in a way that is suitable for both analysis and simulation. The guiding principle is that the bulk theory is posed on four spatial dimensions $\mathbf{X} = (x, y, z, w)$, while brane-facing observables are defined *operationally* by a projection (measurement) map rather than by imposing boundary matching conditions.

2.1 Bulk Geometry and Index Conventions

Coordinates. We work on a $(4 + 1)$ -dimensional spacetime with coordinates

$$x^M = (t, x, y, z, w) \equiv (t, \mathbf{x}, w), \quad \mathbf{X} \equiv (x, y, z, w) \in \mathbb{R}^4, \quad \mathbf{x} \equiv (x, y, z) \in \mathbb{R}^3. \quad (1)$$

The operational brane is centered at $w = 0$. In the full theory w is a bona fide spatial coordinate; “on the brane” will always mean “as measured by the projection defined in Sec. 2.2.”

Indices. We use the following index sets:

$$\begin{aligned}
\text{Bulk spacetime indices:} & \quad M, N, \dots \in \{0, 1, 2, 3, 4\}, \quad 0 \equiv t, \quad 4 \equiv w, \\
\text{Brane spacetime indices:} & \quad \mu, \nu, \dots \in \{0, 1, 2, 3\}, \\
\text{Bulk spatial indices:} & \quad A, B, \dots \in \{1, 2, 3, 4\} \equiv \{x, y, z, w\}, \\
\text{Brane spatial indices:} & \quad a, b, \dots \in \{1, 2, 3\} \equiv \{x, y, z\}.
\end{aligned} \tag{2}$$

We frequently decompose bulk fields into brane and transverse components, e.g.

$$A_M = (A_0, A_a, A_w), \quad \partial_A = (\partial_a, \partial_w). \tag{3}$$

Metric and derivatives. For the electromagnetic sector we use a flat $(4 + 1)$ Minkowski metric with mostly-plus signature,

$$\eta_{MN} = \text{diag}(-1, +1, +1, +1, +1), \tag{4}$$

used for index bookkeeping and for defining the Maxwell kinetic operator. We set the brane light-cone speed to unity in this paper (restore c by $t \mapsto ct$ in η_{MN} and $\partial_t \mapsto c^{-1}\partial_t$). Derivatives are $\partial_M \equiv \partial/\partial x^M$. We define

$$\nabla_3 = (\partial_x, \partial_y, \partial_z), \quad \nabla_3^2 = \partial_x^2 + \partial_y^2 + \partial_z^2, \quad \nabla_4 = (\nabla_3, \partial_w), \quad \nabla_4^2 = \nabla_3^2 + \partial_w^2. \tag{5}$$

The $(4 + 1)$ d'Alembertian is

$$\square_5 \equiv \eta^{MN} \partial_M \partial_N = -\partial_t^2 + \nabla_3^2 + \partial_w^2. \tag{6}$$

Integration measures and boundary terms. We write

$$\int d^4 X \equiv \int_{\mathbb{R}^4} dx dy dz dw, \quad \int d^3 x \equiv \int_{\mathbb{R}^3} dx dy dz. \tag{7}$$

When integrating by parts in w , we will either (i) retain the boundary flux terms explicitly, or (ii) state sufficient decay/localization assumptions for those terms to vanish as $|w| \rightarrow \infty$.

Hybrid relativity (model symmetry). The electromagnetic sector is evolved as a relativistic hyperbolic field theory in $(4 + 1)$ dimensions. The plasma matter sector used in this paper is nonrelativistic (two-fluid or Schrödinger/GNLS type). Thus the combined system is not assumed to be fully $(4 + 1)$ -Lorentz invariant; “covariant” below refers to *gauge covariance* of matter couplings and to the covariant form of the Maxwell sector in flat $(4 + 1)$ spacetime.

2.2 Localization and Projection Operators

A central distinction in this framework is between:

- **Localization:** a weight $Z(w)$ that appears in the *bulk action* and localizes gauge dynamics near $w = 0$.
- **Projection:** a (generally independent) measurement kernel $W(w)$ that defines *brane observables* as weighted averages over w .

Localization profile $Z(w)$. The electromagnetic kinetic term is multiplied by a nonnegative localization profile $Z(w)$ with finite integral

$$Z_{\text{int}} \equiv \int_{-\infty}^{+\infty} dw Z(w) < \infty. \quad (8)$$

A canonical analytic choice is a Gaussian profile,

$$Z(w) = \exp\left(-\frac{w^2}{\lambda^2}\right), \quad Z_{\text{int}} = \lambda\sqrt{\pi}, \quad (9)$$

but none of the definitions in this section require a specific form.

Operational brane projection $W(w)$. We define brane observables by a normalized kernel $W(w)$,

$$\int_{-\infty}^{+\infty} dw W(w) = 1, \quad W(w) \geq 0, \quad W(w) \text{ peaked near } w = 0. \quad (10)$$

For any bulk field $Q(t, \mathbf{x}, w)$, the brane-measured quantity is

$$\overline{Q}(t, \mathbf{x}) \equiv \int_{-\infty}^{+\infty} dw W(w) Q(t, \mathbf{x}, w). \quad (11)$$

In this normalized convention, $\overline{Q}$ has the same units as Q . If one instead wants a strictly dimension-reduced density per brane volume (integrated through the transverse direction), one may use an unnormalized kernel; we adopt (10) because it corresponds to an operational “finite resolution” measurement across w and makes leakage bookkeeping explicit.

Matched kernels (optional). A particularly natural operational choice is the matched kernel

$$W(w) = \frac{Z(w)}{Z_{\text{int}}}, \quad (12)$$

which weights measurements in the same region where the gauge dynamics is localized. We keep W independent in the formalism to separate “how the bulk evolves” (set by Z) from “what a brane observer measures” (set by W).

2.3 Fields and Brane-Observable Decompositions

Gauge field and field strength. The bulk gauge potential is $A_M(t, \mathbf{x}, w)$, with field strength

$$F_{MN} \equiv \partial_M A_N - \partial_N A_M. \quad (13)$$

We frequently separate brane and transverse components:

$$F_{\mu\nu}, \quad F_{\mu w}. \quad (14)$$

Gauge transformations act as $A_M \mapsto A_M + \partial_M \chi$, leaving F_{MN} invariant.

Brane-facing electric and magnetic fields. For later comparison with standard plasma notation, we define brane-facing fields from the W -projected field strength:

$$E_a^{(\text{brane})}(t, \mathbf{x}) \equiv \overline{F_{a0}}(t, \mathbf{x}), \quad (15)$$

$$B_a^{(\text{brane})}(t, \mathbf{x}) \equiv \frac{1}{2} \epsilon_{abc} \overline{F_{bc}}(t, \mathbf{x}), \quad (16)$$

where ϵ_{abc} is the Levi-Civita symbol on the brane with $\epsilon_{xyz} = +1$. The bulk theory also contains physical mixed components that have no analogue in strictly $3 + 1$ Maxwell theory:

$$E_w(t, \mathbf{x}, w) \equiv F_{w0}, \quad C_a(t, \mathbf{x}, w) \equiv F_{aw}. \quad (17)$$

Their brane-observed counterparts are $\overline{E_w}$ and $\overline{C_a}$. In the controlled MHD/Maxwell limit these mixed components are assumed negligible; in the full 4D interaction theory they provide additional force and topology-transport channels.

Plasma observables (multi-species). For each plasma species s (e.g. electrons and ions), let $\rho_s(t, \mathbf{x}, w)$ denote the bulk number density and $j_s^A(t, \mathbf{x}, w)$ the bulk number current in the four spatial directions. Brane-observed densities and brane spatial currents are defined by

$$\rho_{s,\text{brane}}(t, \mathbf{x}) \equiv \overline{\rho_s}(t, \mathbf{x}), \quad j_{s,\text{brane}}^a(t, \mathbf{x}) \equiv \overline{j_s^a}(t, \mathbf{x}). \quad (18)$$

When $\rho_{s,\text{brane}} > 0$, we define a brane-observed species velocity by

$$v_{s,\text{brane}}^a \equiv \frac{j_{s,\text{brane}}^a}{\rho_{s,\text{brane}}}. \quad (19)$$

This is a diagnostic/observable definition; it does *not* imply that the bulk flow is strictly confined to the brane.

2.4 Exact Projected Balance Laws and Leakage Terms

The projection (11) does not, in general, commute with bulk conservation laws. This is the mathematical origin of brane “non-ideality” in a conservative bulk theory.

Exact leakage identity (species continuity). Assume each species satisfies a bulk continuity equation

$$\partial_t \rho_s + \partial_a j_s^a + \partial_w j_s^w = 0. \quad (20)$$

Multiply by $W(w)$ and integrate over w . Using integration by parts in the $\partial_w j_s^w$ term yields the exact projected balance law

$$\partial_t \rho_{s,\text{brane}} + \partial_a j_{s,\text{brane}}^a = - \left[W(w) j_s^w \right]_{w=-\infty}^{w=+\infty} + \int_{-\infty}^{+\infty} dw W'(w) j_s^w(t, \mathbf{x}, w). \quad (21)$$

We define the right-hand side as the leakage/source term

$$S_{\text{leak}}^{(s)}(t, \mathbf{x}) \equiv - \left[W j_s^w \right]_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} dw W'(w) j_s^w. \quad (22)$$

If $W(w)$ is compactly supported or decays sufficiently fast and j_s^w does not grow at infinity, the boundary flux term vanishes and the leakage is carried entirely by the overlap with $W'(w)$. In either case, (21) is an identity: brane densities can change due to transverse transport even when the bulk evolution is exactly conservative.

Interpretation. In the controlled MHD limit, one assumes $S_{\text{leak}}^{(s)} \approx 0$ (no net exchange through w at the measurement scale), and (21) reduces to ordinary 3D continuity on the brane. In the full 4D interaction model, $S_{\text{leak}}^{(s)}$ is retained and becomes a physically meaningful diagnostic: reconnection-like and topology-changing behavior in the brane-facing fields can be reinterpreted as conservative transport into transverse degrees of freedom (j^w , mixed EM components $F_{\mu w}$, and/or mode towers).

Projection versus reduction (summary). We close this section by emphasizing the separation:

- **Reduction** uses $Z_{\text{int}} = \int Z(w)dw$ to obtain effective brane couplings (e.g. μ_0^{eff}) under a stated ansatz (e.g. a brane-dominant zero mode).
- **Projection** uses $W(w)$ to define what is observed on the brane and yields exact open-system identities such as (21).

This distinction is essential for organizing “MHD as a controlled limit” and “beyond-MHD physics as controlled relaxations” within a single bulk theory.

3 4+1D Electromagnetic Sector with Localization

This section defines the electromagnetic (gauge) sector of the model in $4 + 1$ dimensions with a transverse localization profile $Z(w)$. The key features are: (i) the bulk equations are those of Maxwell theory with a weighted kinetic term, (ii) brane-facing Maxwell theory is recovered as a *controlled* reduction with an effective coupling $\mu_{0,\text{eff}}$, and (iii) for Gaussian localization the transverse operator admits a closed-form Hermite (KK) tower whose exchange produces fixed-coefficient Yukawa corrections. These structures provide both a clean derivation of the MHD limit (later) and a computable hierarchy of beyond-MHD degrees of freedom (the A_w sector and transverse-mode excitations).

3.1 Localized Maxwell action and field equations

Action. The bulk gauge potential is $A_M(t, \mathbf{x}, w)$ with field strength

$$F_{MN} \equiv \partial_M A_N - \partial_N A_M, \quad F^{MN} \equiv \eta^{MP} \eta^{NQ} F_{PQ}, \quad (23)$$

where $\eta_{MN} = \text{diag}(-1, +1, +1, +1, +1)$ is the flat bulk metric from Sec. 2. The localized Maxwell action is

$$S_{\text{EM}}[A; Z] = \int dt d^3x dw \left[-\frac{Z(w)}{4\mu_0} F_{MN} F^{MN} - \frac{1}{2\xi\mu_0} (\partial \cdot A)^2 - A_M J^M \right]. \quad (24)$$

The gauge-fixing term (parameter ξ) is included only to make the kinetic operator invertible; all gauge-invariant observables are ξ -independent for conserved currents. The interaction is written $\mathcal{L}_{\text{int}} = -A_M J^M$, so that under a gauge variation $\delta A_M = \partial_M \chi$, $\delta \mathcal{L}_{\text{int}} = -\partial_M (\chi J^M) + \chi \partial_M J^M$; thus gauge invariance of the coupling is equivalent to $\partial_M J^M = 0$ up to boundary terms.

Euler–Lagrange equations. Varying (24) yields the localized Maxwell equations

$$\partial_M (Z(w) F^{MN}) + \frac{1}{\xi} \partial^N (\partial \cdot A) = \mu_0 J^N. \quad (25)$$

These reduce to the usual Maxwell equations when $Z \equiv 1$. Here $Z(w)$ breaks full $4 + 1$ translation/Lorentz symmetry in the w -direction while preserving brane Lorentz covariance in the $(t, \mathbf{x})$ directions.

Bianchi identities (homogeneous Maxwell). By definition of F_{MN} ,

$$\partial_{[L} F_{MN]} = 0, \quad (26)$$

an exact off-shell identity. The brane-facing homogeneous equations follow upon projection to brane observables (Sec. 2.2).

Divergence identity and current conservation. Taking ∂_N of (25) and using antisymmetry of F^{MN} gives

$$\frac{1}{\xi} \square_5 (\partial \cdot A) = \mu_0 \partial_N J^N, \quad \square_5 \equiv \eta^{MN} \partial_M \partial_N = -\partial_t^2 + \nabla_3^2 + \partial_w^2. \quad (27)$$

In Lorenz gauge ($\partial \cdot A = 0$), the field equations enforce bulk current conservation $\partial_N J^N = 0$. Conversely, for conserved sources the gauge-fixing term simply selects a representative in the gauge orbit without affecting gauge-invariant quantities.

3.2 3+1 decomposition and the A_w degree of freedom

We decompose the gauge potential and current into brane and transverse components,

$$A_M = (A_0, A_a, A_w), \quad J^M = (J^0, J^a, J^w), \quad a \in \{x, y, z\}. \quad (28)$$

In addition to the brane-facing electric/magnetic fields, the 4 + 1 theory contains mixed field components with no analogue in strictly 3 + 1 Maxwell theory:

$$E_w \equiv F_{w0} = -\partial_t A_w - \partial_w A_0, \quad C_a \equiv F_{aw} = \partial_a A_w - \partial_w A_a. \quad (29)$$

These enter forces on charged matter through the full Lorentz tensor structure $q F_{iN} u^N$ (later sections). In particular, a dynamical A_w generically drives transverse drift and induces $J^w \neq 0$, providing a natural, conservative channel for brane-observed non-ideality (leakage and topology export) even when the bulk evolution is otherwise “ideal.”

Axial gauge as a convenience (not a physical assumption). A common simplification is axial gauge $A_w = 0$. Since under a gauge transformation $A_w \mapsto A_w + \partial_w \chi$, axial gauge can be reached locally by choosing

$$\chi_{Ax}(t, \mathbf{x}, w) \equiv - \int_0^w A_w(t, \mathbf{x}, w') dw', \quad (30)$$

provided A_w is sufficiently regular in w . The residual gauge freedom then satisfies $\partial_w \chi = 0$, i.e. $\chi = \chi_b(t, \mathbf{x})$, which is the usual 3 + 1 gauge freedom. In this plasma paper we keep the full A_w sector available; axial gauge will be used only when deriving controlled brane limits or when defining the transverse spectral basis.

3.3 Controlled brane Maxwell limit and effective coupling

The localization profile is assumed nonnegative and integrable,

$$Z_{\text{int}} \equiv \int_{-\infty}^{+\infty} Z(w) dw < \infty. \quad (31)$$

A canonical analytic choice (used for explicit spectra below) is Gaussian localization,

$$Z(w) = \exp\left(-\frac{w^2}{\lambda^2}\right), \quad Z_{\text{int}} = \lambda\sqrt{\pi}. \quad (32)$$

Exact w -integrated identity. Consider the ν -components ($\nu \in \{0, 1, 2, 3\}$) of (25) and integrate over w :

$$\partial_\mu \left(\int_{-\infty}^{+\infty} Z F^{\mu\nu} dw \right) + \left[Z F^{w\nu} \right]_{w=-\infty}^{w=+\infty} + \frac{1}{\xi} \partial^\nu \left(\int_{-\infty}^{+\infty} (\partial \cdot A) dw \right) = \mu_0 J_{\text{eff}}^\nu, \quad (33)$$

$$J_{\text{eff}}^\nu(t, \mathbf{x}) \equiv \int_{-\infty}^{+\infty} J^\nu(t, \mathbf{x}, w) dw. \quad (34)$$

For localized field configurations the boundary flux term vanishes. It is then natural to define the Z -weighted brane field strength

$$\langle F^{\mu\nu} \rangle_Z \equiv \frac{1}{Z_{\text{int}}} \int_{-\infty}^{+\infty} Z(w) F^{\mu\nu}(t, \mathbf{x}, w) dw. \quad (35)$$

Dividing (33) by Z_{int} gives the exact integrated brane equation

$$\partial_\mu \langle F^{\mu\nu} \rangle_Z + \frac{1}{\xi Z_{\text{int}}} \partial^\nu \left(\int (\partial \cdot A) dw \right) = \frac{\mu_0}{Z_{\text{int}}} J_{\text{eff}}^\nu. \quad (36)$$

This is *not yet* standard Maxwell, because it still contains the explicit w -integral of the gauge-fixing term and the Z -weighted average of the bulk field strength.

Controlled zero-mode reduction. To recover ordinary $3+1$ Maxwell theory we now take a controlled limit:

1. **Zero-mode dominance:** the brane components are approximately w -independent in the localization region, $A_\mu(t, \mathbf{x}, w) \simeq a_\mu(t, \mathbf{x})$, $\partial_w A_\mu \simeq 0$.
2. **Negligible mixed flux:** $F^{w\nu}$ is negligible and/or sufficiently localized so that $[ZF^{w\nu}]_{-\infty}^{+\infty} = 0$ and $\langle F^{\mu\nu} \rangle_Z \simeq f^{\mu\nu}$ with $f_{\mu\nu} \equiv \partial_\mu a_\nu - \partial_\nu a_\mu$.
3. **No net transverse current in this limit:** the $N = w$ equation is consistent with $J^w \simeq 0$ (see below).

Under these assumptions, (36) reduces to

$$\partial_\mu f^{\mu\nu} + \frac{1}{\xi} \partial^\nu (\partial_\mu a^\mu) = \mu_0^{\text{eff}} J_{\text{eff}}^\nu, \quad \mu_0^{\text{eff}} \equiv \frac{\mu_0}{Z_{\text{int}}}. \quad (37)$$

If the effective brane current is brane-localized (e.g. $J^\mu = J_b^\mu \delta(w)$) then $J_{\text{eff}}^\nu = J_b^\nu$. For Gaussian Z ,

$$\mu_0^{\text{eff}} = \frac{\mu_0}{\lambda \sqrt{\pi}}. \quad (38)$$

In Lorenz gauge on the brane ($\partial_\mu a^\mu = 0$) this becomes the standard inhomogeneous Maxwell equation $\partial_\mu f^{\mu\nu} = \mu_0^{\text{eff}} J_b^\nu$.

Consistency of the transverse component. The $N = w$ component of (25) is

$$\partial_M (Z F^{Mw}) + \frac{1}{\xi} \partial^w (\partial \cdot A) = \mu_0 J^w. \quad (39)$$

In axial gauge $A_w = 0$ with the zero-mode ansatz $\partial_w A_\mu \simeq 0$, one has $F^{\mu w} = -\partial^w A^\mu \simeq 0$ and $\partial^w (\partial \cdot A) \simeq 0$, so (39) reduces to $J^w \simeq 0$. Thus the controlled brane Maxwell limit is internally consistent precisely when transverse current is negligible at the reduction scale. In the full plasma system we *do not* impose $J^w = 0$; instead, J^w becomes a dynamical diagnostic and a source of beyond-MHD terms.

3.4 Mode expansion in the extra dimension

When w -dependence is retained, the localized Maxwell sector contains a normalizable massless mode plus a discrete tower of massive excitations. For Gaussian $Z(w)$ these modes can be computed in closed form and provide a numerically natural basis (Hermite spectral) for large-scale simulations.

Sturm–Liouville eigenproblem. In axial gauge $A_w = 0$ and (for convenience) Lorenz gauge for the brane components, a separated expansion

$$A_\mu(t, \mathbf{x}, w) = \sum_{n \geq 0} a_\mu^{(n)}(t, \mathbf{x}) f_n(w) \quad (40)$$

leads to the transverse Sturm–Liouville problem

$$-\frac{d}{dw} \left(Z(w) \frac{df_n}{dw} \right) = m_n^2 Z(w) f_n(w), \quad (41)$$

with inner product $\langle f, g \rangle_Z = \int Z(w) f(w) g(w) dw$.

Gaussian localization: Hermite spectrum. For $Z(w) = \exp(-w^2/\lambda^2)$, defining $y = w/\lambda$ turns (41) into the Hermite equation. A complete polynomial basis is

$$f_n(w) = H_n\left(\frac{w}{\lambda}\right), \quad m_n^2 = \frac{2n}{\lambda^2}, \quad n = 0, 1, 2, \dots, \quad (42)$$

with $n = 0$ the massless (Maxwell) mode and $n \geq 1$ massive modes.

Norms and brane coupling weights. The natural norm induced by the action is

$$\|f_n\|_Z^2 \equiv \int_{-\infty}^{+\infty} Z(w) f_n(w)^2 dw = \lambda \sqrt{\pi} 2^n n!. \quad (43)$$

For a brane-localized source, each mode is excited in proportion to its value at the brane. A convenient dimensionful coupling weight is

$$c_n \equiv \frac{f_n(0)^2}{\|f_n\|_Z^2}. \quad (44)$$

Since $H_n(0) = 0$ for odd n , odd modes do not couple to brane sources:

$$c_{2m+1} = 0. \quad (45)$$

For even $n = 2m$, using $H_{2m}(0) = (-1)^m (2m)!/m!$,

$$c_{2m} = \frac{1}{\lambda \sqrt{\pi}} \frac{1}{4^m} \binom{2m}{m}, \quad c_0 = \frac{1}{\lambda \sqrt{\pi}} = \frac{1}{Z_{\text{int}}}. \quad (46)$$

Mode-by-mode brane wave operators. For conserved brane sources, each mode propagates on the brane with a covariant massive operator. A convenient summary (in Lorenz gauge) is

$$(\square_4 + m_n^2) a_\mu^{(n)}(t, \mathbf{x}) = \mu_0 c_n J_{b\mu}(t, \mathbf{x}), \quad \square_4 \equiv -\partial_t^2 + \nabla_3^2, \quad (47)$$

with m_n^2 and c_n given by (42) and (44)–(46). The zero mode reproduces Maxwell on the brane with effective coupling $\mu_0^{\text{eff}} = \mu_0 c_0$, while the massive modes are Proca-like degrees of freedom with dispersion $\omega^2 = \mathbf{k}^2 + m_n^2$.

Static brane potential: Coulomb plus Yukawa tower. In the static limit, the brane-to-brane response is a sum of Yukawa propagators. For a point charge q on the brane, the scalar potential on the brane takes the form

$$A_0(r) = \frac{\mu_0 q}{4\pi r} \sum_{n \geq 0} c_n e^{-m_n r}, \quad r \equiv |\mathbf{x}|. \quad (48)$$

Separating the zero mode gives

$$A_0(r) = \frac{\mu_0^{\text{eff}} q}{4\pi r} \left[1 + \sum_{m \geq 1} \frac{c_{2m}}{c_0} e^{-m_{2m} r} \right], \quad m_{2m} = \frac{2\sqrt{m}}{\lambda}, \quad (49)$$

with coefficient ratios

$$\frac{c_{2m}}{c_0} = \frac{1}{4^m} \binom{2m}{m} = 1, \frac{1}{2}, \frac{3}{8}, \frac{5}{16}, \dots \quad (m = 0, 1, 2, 3, \dots). \quad (50)$$

Thus for $r \gg \lambda$ the leading deviation from Coulomb has fixed coefficient $\frac{1}{2}$ and range set by λ :

$$A_0(r) = \frac{\mu_0^{\text{eff}} q}{4\pi r} \left[1 + \frac{1}{2} e^{-2r/\lambda} + \mathcal{O}(e^{-2\sqrt{2}r/\lambda}) \right]. \quad (51)$$

For plasma modeling, λ therefore sets a concrete crossover scale: at separations $r \gg \lambda$ the interaction is effectively Maxwellian, while at $r \sim \lambda$ additional transverse modes and mixed components can contribute. In later sections we show how these same degrees of freedom appear as explicit beyond-MHD correction channels in the brane-facing fluid and induction equations.

4 4+1D Plasma Matter Sector

This section specifies the matter sector used to represent a plasma in the bulk $(t, \mathbf{x}, w)$. The overarching requirement is that the matter variables provide a conserved bulk current J^M suitable for sourcing the localized Maxwell system (Section 3), while also admitting controlled limits that recover standard brane two-fluid plasma theory and MHD (Section 6).

We present two complementary matter realizations:

1. a multi-species nonrelativistic gauge-covariant field theory (Schrödinger/GNLS–Maxwell), which is particularly well-suited for topology change because the fundamental field ψ_s remains regular at $\rho_s \rightarrow 0$ (phase slips are possible);
2. an equivalent kinetic/plasma form (Vlasov–Maxwell in 4 spatial dimensions), which is well-suited for PIC-style massive simulations.

Both realizations lead to the same bulk charge/current bookkeeping and the same brane-projected leakage identities (Section 2.4).

4.1 Choice of Plasma Model: Two-Fluid GNLS or Vlasov

Hybrid relativity (scope). The electromagnetic sector is a relativistic hyperbolic field theory in 4 + 1 dimensions (wave propagation at speed c), while the baseline matter sector is taken to be nonrelativistic. This “Schrödinger–Maxwell” or “two-fluid–Maxwell” hybrid is standard in plasma modeling in the MHD-like regime. If fully relativistic particle dynamics is required, the matter sector can be upgraded (e.g. relativistic kinetic theory), but that extension is not needed for the controlled MHD limit demonstrated here.

Species. We consider S charged species indexed by $s \in \{1, \dots, S\}$, each with mass m_s and charge q_s . In applications, $s = e$ (electrons) and $s = i$ (ions) are the minimal choice.

Option A (field-based): multi-species Schrödinger/GNLS–Maxwell

Gauge-covariant derivatives. For each species,

$$D_t^{(s)} \equiv \partial_t + \frac{iq_s}{\hbar} A_0, \quad D_A^{(s)} \equiv \partial_A - \frac{iq_s}{\hbar} A_A, \quad A \in \{x, y, z, w\}. \quad (52)$$

Matter action (nonrelativistic, gauge-covariant). A minimal multi-species matter action consistent with the unified model style is

$$S_{\text{matt}}[\{\psi_s\}, A] = \sum_{s=1}^S \int dt d^3x dw \left[\frac{i\hbar}{2} \left(\psi_s^* D_t^{(s)} \psi_s - \psi_s (D_t^{(s)} \psi_s)^* \right) - \frac{\hbar^2}{2m_s} (D_A^{(s)} \psi_s)^* (D_A^{(s)} \psi_s) - V_{\text{conf},s}(\mathbf{x}, w) |\psi_s|^2 - U_s(|\psi_s|^2) \right], \quad (53)$$

where $V_{\text{conf},s}$ is an optional confinement/geometry potential and $U_s(\rho_s)$ is an equation-of-state (EOS) potential encoding pressure/enthalpy effects in the hydrodynamic limit. The matter model is *gauge covariant* by construction, and gauge invariance of the interaction term implies current conservation (see below).

Field equations (GNLS). Variation of (53) yields, for each species,

$$i\hbar D_t^{(s)} \psi_s = \left[-\frac{\hbar^2}{2m_s} D_A^{(s)} D_A^{(s)} + V_{\text{conf},s} + h_s(\rho_s) \right] \psi_s, \quad \rho_s \equiv |\psi_s|^2, \quad h_s(\rho_s) \equiv \frac{dU_s}{d\rho_s}. \quad (54)$$

A frequently used stiff-polytrope choice compatible with earlier unified-model fixes is $U_s(\rho_s) = \frac{K_s}{4} \rho_s^5$, giving $h_s(\rho_s) = \frac{5K_s}{4} \rho_s^4$; however we keep U_s general in this section because plasma closures may vary by regime.

Option B (kinetic): 4D Vlasov–Maxwell

As an alternative (or complement) to the field-based realization, each species can be described by a phase-space distribution $f_s(t, \mathbf{x}, w; \mathbf{v})$ where $\mathbf{v} \in \mathbb{R}^4$ is the bulk spatial velocity (v_x, v_y, v_z, v_w) . The collisionless Vlasov equation in the bulk is

$$\partial_t f_s + v^A \partial_A f_s + \frac{q_s}{m_s} (E_A + v^B F_{AB}) \partial_{v_A} f_s = 0, \quad A, B \in \{x, y, z, w\}. \quad (55)$$

Collisions (when desired) can be included through an additional operator $C_s[f]$ on the right-hand side (e.g. BGK, Landau, or effective friction models), provided the chosen operator preserves overall charge conservation.

Bulk moments define density and current:

$$\rho_s(t, \mathbf{x}, w) = \int_{\mathbb{R}^4} f_s d^4v, \quad j_s^A(t, \mathbf{x}, w) = \int_{\mathbb{R}^4} v^A f_s d^4v. \quad (56)$$

4.2 Species Variables, Currents, and Conservation

Number density and number current. In both realizations (GNLS or Vlasov), the species number density is denoted $\rho_s(t, \mathbf{x}, w)$. The associated bulk number current is denoted $j_s^A(t, \mathbf{x}, w)$. For GNLS,

$$j_s^A = \frac{\hbar}{m_s} \text{Im} \left(\psi_s^* D_A^{(s)} \psi_s \right), \quad A \in \{x, y, z, w\}. \quad (57)$$

For Vlasov, j_s^A is the velocity moment in (56).

Exact bulk continuity. Each species satisfies an exact bulk continuity equation,

$$\partial_t \rho_s + \partial_A j_s^A = 0. \quad (58)$$

This identity guarantees gauge invariance of the coupling to the Maxwell sector.

Charge density and charge current (Maxwell sources). We define the species 5-current J_s^M by

$$J_s^0 \equiv q_s \rho_s, \quad J_s^A \equiv q_s j_s^A, \quad (59)$$

and the total conserved source is

$$J^M \equiv J_{\text{ext}}^M + \sum_{s=1}^S J_s^M, \quad \partial_M J^M = 0. \quad (60)$$

Here J_{ext}^M is reserved for additional sources not represented by the chosen matter closure (e.g. externally imposed drive currents or explicit defect worldlines in hybrid simulations).

Projected (brane) continuity with leakage. Applying the brane projection map $\bar{\cdot}$ from Sec. 2.2 to (58) yields the exact open-system identity

$$\partial_t \rho_{s,\text{brane}} + \partial_a j_{s,\text{brane}}^a = S_{\text{leak}}^{(s)}(t, \mathbf{x}), \quad (61)$$

with leakage source $S_{\text{leak}}^{(s)}$ given explicitly in (22). This term vanishes only under additional controlled assumptions (e.g. negligible transverse flux j_s^w at the measurement scale).

4.3 Forces and Canonical Momentum

Hydrodynamic (Madelung) representation and canonical momentum

For the field-based model, write the species wavefunction in amplitude–phase form (Madelung transform)

$$\psi_s(t, \mathbf{x}, w) = \sqrt{\rho_s(t, \mathbf{x}, w)} \exp(i\theta_s(t, \mathbf{x}, w)). \quad (62)$$

Define the bulk species velocity $\mathbf{v}_s \in \mathbb{R}^4$ by $j_s^A = \rho_s v_s^A$. Then

$$v_s^A = \frac{\hbar}{m_s} \left(\partial^A \theta_s - \frac{q_s}{\hbar} A^A \right). \quad (63)$$

The corresponding canonical momentum is

$$p_{s,A}^{\text{can}} \equiv \hbar \partial_A \theta_s, \quad p_{s,A}^{\text{kin}} \equiv m_s v_{s,A}, \quad p_{s,A}^{\text{can}} = p_{s,A}^{\text{kin}} + q_s A_A. \quad (64)$$

This identity is important computationally: evolving ψ_s (or equivalently θ_s and ρ_s) evolves canonical momentum in a way that remains well-defined even when $\rho_s \rightarrow 0$, whereas raw velocity $\mathbf{v}_s$ can become singular in such regions. This feature is one reason the field-based system is attractive for reconnection/topology-change studies.

Euler-like momentum equation (bulk 4D space)

Inserting (62) into the GNLS equation (54) yields the exact continuity equation (58) and an Euler-like momentum equation for each species:

$$m_s (\partial_t + v_s^B \partial_B) v_{s,A} = q_s (E_A + v_s^B F_{AB}) - \partial_A (V_{\text{conf},s} + h_s(\rho_s) + Q_s(\rho_s)), \quad (65)$$

where

$$E_A \equiv F_{A0} = -\partial_t A_A - \partial_A A_0, \quad F_{AB} = \partial_A A_B - \partial_B A_A, \quad (66)$$

and the quantum (dispersive) potential is

$$Q_s(\rho_s) \equiv -\frac{\hbar^2}{2m_s} \frac{\nabla_4^2 \sqrt{\rho_s}}{\sqrt{\rho_s}}. \quad (67)$$

The classical two-fluid limit is obtained by dropping Q_s (or taking a controlled $\hbar \rightarrow 0$ limit) while retaining the barotropic closure encoded in $h_s(\rho_s)$.

Transverse forces and the role of A_w . Equation (65) holds for all bulk spatial components $A \in \{x, y, z, w\}$. In particular, the $A = w$ component reads

$$m_s (\partial_t + v_s^B \partial_B) v_{s,w} = q_s (E_w + v_s^a F_{wa}) - \partial_w (V_{\text{conf},s} + h_s + Q_s), \quad (68)$$

with $E_w = F_{w0} = -\partial_t A_w - \partial_w A_0$. Thus a dynamical A_w (and/or mixed components F_{wa}) generically drives transverse drift $v_{s,w} \neq 0$, which in turn produces $j_s^w \neq 0$ and therefore nonzero projected leakage $S_{\text{leak}}^{(s)}$ on the brane. This is one of the primary 4D interaction channels that will appear as beyond-MHD corrections in Section 7.

Collisions and effective resistivity (optional). When a resistive or collisional regime is desired, one may include momentum-exchange terms in the species momentum equations (or collision operators in Vlasov) that preserve overall charge conservation. For example, an interspecies drag term of the form $-m_s \nu_{ss'} (v_s^A - v_{s'}^A)$ can be added to the right-hand side of (65); in the controlled brane reduction such terms generate standard resistive contributions to Ohm's law. In the present framework, however, non-ideal behavior may also arise *without* such dissipative closures via brane–bulk exchange and transverse-mode excitation.

Useful kinematic identity (canonical vorticity). From (63) one obtains, away from phase singularities,

$$\partial_A v_{s,B} - \partial_B v_{s,A} = -\frac{q_s}{m_s} F_{AB}. \quad (69)$$

This “London-type” identity states that canonical vorticity is locked to gauge flux as a bulk geometric constraint; its brane-facing implications for topology, reconnection, and projected non-ideality are developed in Section ??.

5 Full Coupled 4+1D Plasma–EM System

We now collect the coupled bulk equations that define the simulation target: a localized 4 + 1 Maxwell sector (Section 3) sourced by a multi-species plasma matter sector (Section 4). The resulting system is conservative in the bulk (up to stated boundary/localization conditions) and admits controlled reductions to standard 3 + 1 Maxwell theory and MHD on the brane (Section 6). Throughout, the brane is not a boundary of the PDE domain; brane-facing quantities are defined by the projection operator $\bar{\cdot}$ (Section 2.2) and may exhibit explicit exchange (leakage) terms even when bulk dynamics is conservative.

5.1 Coupled action and equations of motion

Total action (bulk). A convenient parent formulation is the sum of the localized Maxwell action (24) and a multi-species matter action (field-based option shown here; kinetic option below):

$$S_{\text{tot}} = S_{\text{EM}}[A; Z] + \sum_{s=1}^S S_{\text{matt},s}[\psi_s, A] + S_{\text{ext}}[A], \quad (70)$$

where S_{ext} allows additional prescribed sources (drive currents, background fields, etc.). The localization profile $Z(w)$ is fixed (time independent) and integrable.

Maxwell equation with localization. Variation with respect to A_N yields

$$\partial_M(Z(w)F^{MN}) + \frac{1}{\xi} \partial^N(\partial \cdot A) = \mu_0 J_{\text{tot}}^N, \quad J_{\text{tot}}^N \equiv J_{\text{ext}}^N + \sum_{s=1}^S J_s^N, \quad (71)$$

with the exact Bianchi identity $\partial_{[L}F_{MN]} = 0$. For conserved sources $\partial_N J_{\text{tot}}^N = 0$, the gauge-fixing term does not affect gauge-invariant observables.

Matter evolution: field-based (GNLS) option. For each species s , the gauge-covariant derivatives are

$$D_t^{(s)} \equiv \partial_t + \frac{iq_s}{\hbar} A_0, \quad D_A^{(s)} \equiv \partial_A - \frac{iq_s}{\hbar} A_A, \quad A \in \{x, y, z, w\}, \quad (72)$$

and the species field equation is

$$i\hbar D_t^{(s)} \psi_s = \left[-\frac{\hbar^2}{2m_s} D_A^{(s)} D_A^{(s)} + V_{\text{conf},s}(\mathbf{x}, w) + h_s(\rho_s) \right] \psi_s, \quad \rho_s \equiv |\psi_s|^2, \quad h_s(\rho_s) \equiv \frac{dU_s}{d\rho_s}. \quad (73)$$

The confinement potential $V_{\text{conf},s}$ may be used to localize plasma species near the operational brane or near defect/throat geometries; for an unconstrained bulk plasma one sets $V_{\text{conf},s} \equiv 0$.

Matter evolution: kinetic (Vlasov) option. Alternatively, each species may be represented by a bulk phase-space distribution $f_s(t, \mathbf{x}, w; \mathbf{v})$ with $\mathbf{v} \in \mathbb{R}^4$, evolving via

$$\partial_t f_s + v^A \partial_A f_s + \frac{q_s}{m_s} (E_A + v^B F_{AB}) \partial_{v_A} f_s = C_s[f], \quad (74)$$

where $C_s[f]$ is an optional collision operator chosen to preserve overall charge conservation. The electromagnetic fields are defined from F_{MN} by $E_A \equiv F_{A0}$ and F_{AB} as in (66).

Species sources (currents). In the GNLS representation, the bulk number current is

$$j_s^A = \frac{\hbar}{m_s} \text{Im}(\psi_s^* D_A^{(s)} \psi_s), \quad (75)$$

and the species 5-current sourcing Maxwell is

$$J_s^0 \equiv q_s \rho_s, \quad J_s^A \equiv q_s j_s^A. \quad (76)$$

In the kinetic representation, ρ_s and j_s^A are the moments (56). In either case, each species satisfies exact bulk continuity

$$\partial_t \rho_s + \partial_A j_s^A = 0 \quad \Rightarrow \quad \partial_M J_s^M = 0, \quad (77)$$

and therefore $\partial_M J_{\text{tot}}^M = 0$ when $\partial_M J_{\text{ext}}^M = 0$.

Simulation target (compact statement). For later reference, the coupled bulk system to be evolved can be written as:

$$\begin{aligned} & \partial_M(Z(w)F^{MN}) + \frac{1}{\xi} \partial^N(\partial \cdot A) = \mu_0 \left(J_{\text{ext}}^N + \sum_s J_s^N \right), \\ & \text{either } i\hbar D_t^{(s)} \psi_s = \left[-\frac{\hbar^2}{2m_s} D_A^{(s)} D_A^{(s)} + V_{\text{conf},s} + h_s(\rho_s) \right] \psi_s \quad (\text{field/GNLS}), \\ & \text{or } \partial_t f_s + v^A \partial_A f_s + \frac{q_s}{m_s} (E_A + v^B F_{AB}) \partial_{v_A} f_s = C_s[f] \quad (\text{kinetic/Vlasov}), \\ & J_s^0 = q_s \rho_s, \quad J_s^A = q_s j_s^A, \quad \partial_M J_s^M = 0, \quad \partial_{[L} F_{MN]} = 0. \end{aligned} \quad (78)$$

5.2 Bulk constraints, gauge choices, and well-posedness

Constraints. When evolving potentials A_M , the homogeneous Maxwell equations (26) are automatically satisfied. The inhomogeneous equations (71) contain constraint content in the $N = 0$ component (a Gauss-type law) and evolution content in the $N = A$ components (Ampère-type laws). The gauge-fixing term in (71) promotes the system to a manifestly hyperbolic form suitable for time-domain simulation.

Gauge handling. We do not assume $A_w = 0$ in the full theory; the A_w sector and the mixed components $F_{\mu w}$ are retained as physical degrees of freedom and are a key source of beyond-MHD corrections on the brane (Section 7). For analysis and for transverse mode decompositions it can be convenient to impose axial gauge $A_w = 0$ with residual brane gauge freedom $\chi_b(t, \mathbf{x})$, but this is treated as a *controlled* simplification rather than a defining assumption.

Localization and boundary conditions. Because $Z(w)$ breaks translation invariance in the w -direction, conserved Noether momenta along w are not expected in general. Time translation symmetry is preserved (Z is time independent), so the system admits a conserved total energy under suitable boundary/localization conditions. In practice, one assumes either (i) sufficiently fast decay of fields/currents as $|w| \rightarrow \infty$, or (ii) effective compact support in w as realized by spectral/Hermite truncation or numerical domain truncation (Section 9).

5.3 Energy ledger and conservation (bulk)

A major advantage of the action-based formulation is that it provides a clean energy ledger for verification and diagnostics in simulations. Here we record a gauge-invariant bulk energy functional and its exchange structure between matter and fields.

Electromagnetic energy density (localized). Define the bulk electric-type and magnetic-type components by

$$E_A \equiv F_{A0}, \quad F_{AB} = \partial_A A_B - \partial_B A_A, \quad A, B \in \{x, y, z, w\}. \quad (79)$$

A convenient gauge-invariant electromagnetic energy density is

$$u_{\text{EM}} \equiv \frac{Z(w)}{2\mu_0} \left(E_A E_A + \frac{1}{2} F_{AB} F_{AB} \right). \quad (80)$$

In terms of brane and mixed components, this contains contributions from the usual brane fields F_{ab} as well as from the mixed sector F_{aw} and the transverse electric component $E_w = F_{w0}$.

Matter energy density (field-based option). For the GNLS realization, a gauge-invariant matter energy density for each species is

$$u_s \equiv \frac{\hbar^2}{2m_s} (D_A^{(s)} \psi_s)^* (D_A^{(s)} \psi_s) + V_{\text{conf},s} \rho_s + U_s(\rho_s), \quad (81)$$

so that the total matter energy density is $u_{\text{matt}} \equiv \sum_s u_s$. (When desired, (81) can be rewritten in hydrodynamic variables as kinetic + internal + dispersive/quantum contributions via the Madelung map.)

Total energy and exchange structure. Define the total bulk energy

$$\mathcal{E}_{\text{tot}}(t) \equiv \int_{\mathbb{R}^3} d^3x \int_{\mathbb{R}} dw (u_{\text{EM}} + u_{\text{matt}}), \quad (82)$$

with u_{EM} from (80) and u_{matt} from (81) (or from kinetic moments in the Vlasov realization). Under standard localization/boundary assumptions and for conservative collision models, $\mathcal{E}_{\text{tot}}$ is conserved. Locally, the EM–matter coupling exchanges energy through the work density $J^A E_A$, which includes the transverse channel $J^w E_w$ whenever the scalar-photon sector is active:

$$\partial_t u_{\text{EM}} + \partial_A S_{\text{EM}}^A = -J_{\text{tot}}^A E_A, \quad \partial_t u_{\text{matt}} + \partial_A S_{\text{matt}}^A = +J_{\text{tot}}^A E_A, \quad (83)$$

for suitable (model-dependent) fluxes S_{EM}^A and S_{matt}^A . A convenient covariant generalization of the Poynting flux is

$$S_{\text{EM}}^A \equiv \frac{Z(w)}{\mu_0} E_B F_{BA}, \quad (84)$$

which reduces to the usual $\mathbf{E} \times \mathbf{B}$ form when restricted to brane components.

Why the energy ledger matters for reconnection. In standard resistive MHD, reconnection is often diagnosed by conversion of magnetic energy into internal energy through ηJ^2 . In the present model, reconnection-like behavior in brane observables can occur through conservative transfer of energy into transverse degrees of freedom: A_w , E_w , mixed components F_{aw} , and/or higher transverse modes (Section D). The energy ledger (82)–(83) provides an unambiguous way to distinguish genuine dissipation (collisions) from topology/energy export into bulk channels.

6 Controlled Reduction: 4D $\rightarrow$ 3+1 Maxwell $\rightarrow$ Two-Fluid $\rightarrow$ MHD

This section establishes that standard 3+1 Maxwell theory and classical MHD arise as *controlled limits* of the full 4 + 1 bulk plasma–EM system (Section 5). The reductions proceed in three stages:

$$\begin{aligned} & \text{(bulk localized Maxwell + bulk plasma)} \\ & \implies \text{(effective brane Maxwell + brane two-fluid)} \\ & \implies \text{(single-fluid MHD)}. \end{aligned} \quad (85)$$

The purpose of these reductions is twofold: (i) to show that MHD is recoverable and therefore is not contradicted by the parent theory, and (ii) to expose precisely which assumptions suppress the new 4D interaction channels (leakage J^w , scalar-photon sector A_w , and transverse mode excitation) that will be reinstated in Section 7.

6.1 Controlled assumptions and ordering

We group assumptions by reduction step. Each group defines an approximation regime in which the subsequent reduced equations are valid.

Step 1: localized 4+1 Maxwell $\rightarrow$ effective 3+1 Maxwell

We adopt the brane-dominant zero-mode approximation on the brane components:

1. Zero-mode dominance in the localization region:

$$A_\mu(t, \mathbf{x}, w) \simeq a_\mu(t, \mathbf{x}), \quad \partial_w A_\mu \simeq 0, \quad (86)$$

for $|w|$ in the support of $Z(w)$.

2. Negligible mixed flux in the controlled limit:

$$F_{\mu w} \simeq 0, \quad \text{equivalently} \quad E_w = F_{w0} \simeq 0, \quad C_a = F_{aw} \simeq 0, \quad (87)$$

so that the brane-facing dynamics is dominated by $F_{\mu\nu}$.

3. No net transverse current at the reduction scale:

$$J^w \simeq 0. \quad (88)$$

(When this fails, brane observables become open-system variables with explicit exchange; see Section 7.)

4. Gauge choice (optional): axial gauge $A_w = 0$ may be imposed for convenience, but is not required for the full theory.

Under these assumptions, the brane components reduce to standard 3+1 Maxwell equations with an effective coupling

$$\mu_0^{\text{eff}} \equiv \frac{\mu_0}{Z_{\text{int}}}, \quad Z_{\text{int}} \equiv \int_{-\infty}^{+\infty} Z(w) dw. \quad (89)$$

If desired, define the corresponding effective permittivity by $\epsilon_0^{\text{eff}} \mu_0^{\text{eff}} = c^{-2}$.

Step 2: bulk plasma $\rightarrow$ brane two-fluid

To obtain standard brane plasma equations, we assume the species are effectively brane-supported at the measurement scale:

$$j_s^w \simeq 0 \quad \Rightarrow \quad S_{\text{leak}}^{(s)} \simeq 0, \quad (90)$$

so that projected continuity reduces to ordinary 3D continuity. We further assume a classical fluid closure for each species, i.e. drop dispersive/quantum terms (Q_s) when starting from the GNLS realization.

Step 3: brane two-fluid $\rightarrow$ MHD

The standard MHD reduction is obtained by the familiar controlled assumptions:

1. **Quasi-neutrality:** $n_i \simeq n_e \equiv n$.
2. **Electron inertia suppression:** $m_e/m_i \ll 1$ and electron inertia terms are negligible in Ohm's law.
3. **Low-frequency / long-scale:** optionally neglect displacement current in Ampère's law (Darwin/MHD ordering).

Relaxing any of these yields standard extended-MHD corrections (Hall, electron pressure, electron inertia, resistivity). In the present framework, additional beyond-MHD corrections also arise when the Step-1 and Step-2 assumptions are relaxed (Section 7).

6.2 Two-fluid algebra on the brane

Under the Step-1 and Step-2 assumptions, we work with brane 3-vectors $\mathbf{E}(t, \mathbf{x})$, $\mathbf{B}(t, \mathbf{x})$, and brane currents. The effective Maxwell system is

$$\nabla_3 \cdot \mathbf{B} = 0, \quad \nabla_3 \times \mathbf{E} = -\partial_t \mathbf{B}, \quad (91)$$

$$\nabla_3 \cdot \mathbf{E} = \frac{\rho_q}{\epsilon_0^{\text{eff}}}, \quad \nabla_3 \times \mathbf{B} - \frac{1}{c^2} \partial_t \mathbf{E} = \mu_0^{\text{eff}} \mathbf{J}, \quad (92)$$

with charge density ρ_q and current density $\mathbf{J}$ defined on the brane.

Brane two-fluid equations. For species $s \in \{i, e\}$ with charge $q_i = +e$, $q_e = -e$, we take

$$\partial_t n_s + \nabla_3 \cdot (n_s \mathbf{v}_s) = 0, \quad (93)$$

$$m_s n_s (\partial_t + \mathbf{v}_s \cdot \nabla_3) \mathbf{v}_s = q_s n_s (\mathbf{E} + \mathbf{v}_s \times \mathbf{B}) - \nabla_3 p_s + \mathbf{R}_s, \quad (94)$$

where p_s is species pressure and $\mathbf{R}_s$ represents optional collisional momentum exchange (e.g. electron-ion drag), with $\sum_s \mathbf{R}_s = 0$.

Single-fluid variables. Define the (approximate) mass density, center-of-mass velocity, and current:

$$\rho \equiv m_i n_i + m_e n_e \simeq (m_i + m_e) n, \quad \mathbf{v} \equiv \frac{m_i n_i \mathbf{v}_i + m_e n_e \mathbf{v}_e}{\rho}, \quad \mathbf{J} \equiv en(\mathbf{v}_i - \mathbf{v}_e). \quad (95)$$

Under quasi-neutrality and singly-ionized ions, these relations can be inverted to express species velocities in terms of $\mathbf{v}$ and $\mathbf{J}$:

$$\mathbf{v}_i = \mathbf{v} + \frac{m_e}{m_i + m_e} \frac{\mathbf{J}}{en}, \quad \mathbf{v}_e = \mathbf{v} - \frac{m_i}{m_i + m_e} \frac{\mathbf{J}}{en}. \quad (96)$$

In the electron-inertia-suppressed limit $m_e/m_i \rightarrow 0$,

$$\mathbf{v}_i \rightarrow \mathbf{v}, \quad \mathbf{v}_e \rightarrow \mathbf{v} - \frac{\mathbf{J}}{en}. \quad (97)$$

Generalized Ohm's law (electron momentum). Start from the electron momentum equation (94). Neglecting electron inertia and (optionally) retaining collisions via $\mathbf{R}_e$, one obtains

$$0 = -en(\mathbf{E} + \mathbf{v}_e \times \mathbf{B}) - \nabla_3 p_e + \mathbf{R}_e. \quad (98)$$

Solving for $\mathbf{E}$ and substituting $\mathbf{v}_e$ from (97) yields

$$\mathbf{E} + \mathbf{v} \times \mathbf{B} = \underbrace{\frac{\mathbf{J} \times \mathbf{B}}{en}}_{\text{Hall}} - \underbrace{\frac{1}{en} \nabla_3 p_e}_{\text{electron pressure}} + \underbrace{\frac{\mathbf{R}_e}{en}}_{\text{collisions/resistivity}}. \quad (99)$$

The standard resistive closure corresponds to $\mathbf{R}_e \approx en \eta \mathbf{J}$, giving $\mathbf{E} + \mathbf{v} \times \mathbf{B} = (\mathbf{J} \times \mathbf{B})/(en) - (\nabla_3 p_e)/(en) + \eta \mathbf{J}$. The *ideal MHD* limit is the controlled sub-limit in which the right-hand side of (99) is negligible:

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B}. \quad (100)$$

Single-fluid momentum equation. Adding the ion and electron momentum equations (94) and using $\sum_s \mathbf{R}_s = 0$ yields the single-fluid momentum balance

$$\rho (\partial_t + \mathbf{v} \cdot \nabla_3) \mathbf{v} = \mathbf{J} \times \mathbf{B} - \nabla_3 p, \quad p \equiv p_i + p_e, \quad (101)$$

up to the usual higher-order corrections (electron inertia, viscosity, anisotropic pressure) that are included in extended-MHD closures if required.

6.3 Induction equation and ideal MHD limit

Faraday's law (91) gives

$$\partial_t \mathbf{B} = -\nabla_3 \times \mathbf{E}. \quad (102)$$

Inserting the generalized Ohm law (99) yields the generalized induction equation

$$\partial_t \mathbf{B} = \nabla_3 \times (\mathbf{v} \times \mathbf{B}) - \nabla_3 \times \left(\frac{\mathbf{J} \times \mathbf{B}}{en} \right) + \nabla_3 \times \left(\frac{1}{en} \nabla_3 p_e \right) - \nabla_3 \times \left(\frac{\mathbf{R}_e}{en} \right). \quad (103)$$

In the ideal MHD limit (100), this reduces to the standard ideal induction equation

$$\partial_t \mathbf{B} = \nabla_3 \times (\mathbf{v} \times \mathbf{B}). \quad (104)$$

Low-frequency Ampère closure (optional). In many MHD regimes one further neglects displacement current, reducing (92) to

$$\nabla_3 \times \mathbf{B} \simeq \mu_0^{\text{eff}} \mathbf{J}, \quad (105)$$

so that $\mathbf{J}$ may be eliminated in favor of $\mathbf{B}$ in (101) and (103). This closure is controlled by $\omega/(ck)$ and is not essential to the logical derivation of ideal MHD from (100).

Summary: standard MHD system. A common ideal-MHD closure in this notation consists of

$$\partial_t \rho + \nabla_3 \cdot (\rho \mathbf{v}) = 0, \quad (106)$$

$$\rho (\partial_t + \mathbf{v} \cdot \nabla_3) \mathbf{v} = \mathbf{J} \times \mathbf{B} - \nabla_3 p, \quad (107)$$

$$\partial_t \mathbf{B} = \nabla_3 \times (\mathbf{v} \times \mathbf{B}), \quad \nabla_3 \cdot \mathbf{B} = 0, \quad (108)$$

supplemented by an equation of state and (optionally) (105) to relate $\mathbf{J}$ and $\mathbf{B}$. The only modification relative to standard SI MHD is that the effective coupling μ_0 is replaced by $\mu_0^{\text{eff}} = \mu_0/Z_{\text{int}}$, inherited from the localized 4 + 1 Maxwell sector.

6.4 Symbolic verification (WL referee suite)

The two-fluid algebra leading to (96)–(99) and the ideal induction equation (104) was verified symbolically using a Wolfram Language referee script (`4d_to_mhd_reduction.wl`) maintained alongside the project sources. The script:

1. encodes the controlled brane/zero-mode assumptions for the EM sector and records $\mu_0^{\text{eff}} = \mu_0/Z_{\text{int}}$;
2. solves the linear system relating $(\mathbf{v}_i, \mathbf{v}_e)$ to $(\mathbf{v}, \mathbf{J})$;
3. derives the inertialess-electron generalized Ohm law and expresses it in the standard form $\mathbf{E} + \mathbf{v} \times \mathbf{B} = \dots$;
4. combines Faraday’s law with the ideal Ohm sub-limit to yield $\partial_t \mathbf{B} = \nabla_3 \times (\mathbf{v} \times \mathbf{B})$.

We do not include the script source code in this paper; it is intended as a machine-checkable supplement to the human-readable derivation above.

Where the suppressed 4D terms re-enter. The script also records, as an explicit reminder, the three primary ways in which the controlled MHD reduction can fail in the parent theory: (i) brane leakage in continuity/momentum via j^w (and thus $S_{\text{leak}} \neq 0$), (ii) additional force channels mediated by the A_w sector and mixed components $F_{\mu w}$, and (iii) finite-localization transverse-mode (KK/Hermite) corrections that modify effective propagation and storage of field energy. These are developed as beyond-MHD extensions in Section 7.

7 Beyond-MHD Extensions from 4D Interactions

Section 6 established that standard MHD arises as a *controlled* brane/zero-mode limit of the full 4 + 1D plasma–EM system. The point of that exercise was not merely to recover known equations, but to identify precisely which assumptions suppress the degrees of freedom that are absent in strictly 3 + 1 electromagnetism. In this section we relax those assumptions in a controlled way and classify the resulting *beyond-MHD* correction channels.

The organizing principle is simple:

The bulk evolution is conservative and local in $4+1D$; the brane-facing system is an open subsystem defined by an operational projection map $\overline{(\cdot)}$.

Consequently, “non-ideal” brane behavior can arise from (i) explicit exchange with the transverse direction w and (ii) projection covariances (unresolved transverse structure), even when no ad hoc dissipation is introduced in the bulk.

7.1 What changes when we move beyond the controlled limit

The controlled MHD reduction relied on suppressing transverse structure and mixed EM components:

$$\partial_w(\cdot) \approx 0, \quad A_w \approx 0, \quad F_{\mu w} \approx 0, \quad J^w \approx 0, \quad (\text{zero-mode dominance}). \quad (109)$$

Beyond MHD, we explicitly *allow* these quantities to be non-negligible. The resulting new physics falls into three coupled classes:

1. **Mixed-sector electromagnetism (scalar-photon channel).** The component A_w is a genuine dynamical degree of freedom in $4+1D$ gauge theory. It generates the transverse electric field

$$E_w \equiv F_{w0} = -\partial_t A_w - \partial_w A_0, \quad (110)$$

and together with the mixed components

$$C_a \equiv F_{aw} = \partial_a A_w - \partial_w A_a, \quad a \in \{x, y, z\}, \quad (111)$$

it introduces new force channels that are invisible in strictly $3+1$ Maxwell theory.

2. **Explicit brane–bulk exchange (leakage).** Species can drift in w , producing $j_s^w \neq 0$ and therefore a projected (open-system) continuity law on the brane. Exchange with w is not a numerical artifact; it is a physical degree of freedom of the parent theory.
3. **Finite-localization / transverse-mode corrections.** When the localization profile $Z(w)$ is integrable (Gaussian “soft wall” in particular), the brane-facing EM sector contains a tower of transverse modes with masses $m_n^2 \propto n/\lambda^2$ (Appendix D). These modes provide additional storage/transport channels for energy and topology. In the static limit, they generate Yukawa corrections to Coulomb interactions at scales $r \sim \lambda$ (Appendix D).

In the remainder of this section we write the brane-facing balances in a form that separates *standard extended-MHD terms* (Hall/pressure/inertia/resistivity) from the new *$4D$ interaction terms* (mixed-sector forces, leakage, and projection covariances).

7.2 Open-system brane balances: leakage and projection covariances

The brane-observed quantity $\overline{Q}(t, \mathbf{x})$ is defined by the operational projection kernel $W(w)$ as

$$\overline{Q}(t, \mathbf{x}) \equiv \int_{-\infty}^{+\infty} dw W(w) Q(t, \mathbf{x}, w), \quad \int dw W(w) = 1. \quad (112)$$

Because products do not commute with projection, it is convenient to define the covariance operator

$$\text{Cov}_W[Q, R] \equiv \overline{QR} - \overline{Q} \overline{R}. \quad (113)$$

These covariances are *exact* and represent unresolved transverse structure (higher w -modes in the Hermite/KK language, or fine structure on a direct w -grid).

Species continuity with leakage

Let $\rho_s(t, \mathbf{x}, w)$ and $j_s^A(t, \mathbf{x}, w)$ be the bulk number density and number current for species s , with $A \in \{x, y, z, w\}$. The bulk continuity equation is

$$\partial_t \rho_s + \partial_a j_s^a + \partial_w j_s^w = 0. \quad (114)$$

Projecting with $W(w)$ yields a brane-facing open-system balance:

$$\partial_t \overline{\rho_s} + \partial_a \overline{j_s^a} = S_{\text{leak}}^{(s)}, \quad (115)$$

where the leakage/source term is

$$S_{\text{leak}}^{(s)} \equiv - \int dw W(w) \partial_w j_s^w = - \left[W j_s^w \right]_{-\infty}^{+\infty} + \int dw W'(w) j_s^w. \quad (116)$$

When the boundary term vanishes, $S_{\text{leak}}^{(s)} = \int dw W'(w) j_s^w$. This term is measured directly in both simulation architectures: direct w -grid codes compute it from j_s^w ; 3D $\times$ modes codes compute it from reconstructed $j_s^w(t, \mathbf{x}, w_q)$ at quadrature nodes.

Brane-facing momentum and EMF covariances

Whenever a brane-facing equation contains a projected product, covariances appear. The most important example for magnetic topology is the projected electromotive term $\mathbf{v} \times \mathbf{B}$, which differs from $\overline{\mathbf{v}} \times \overline{\mathbf{B}}$ by an *exact covariance*:

$$\overline{\mathbf{v} \times \mathbf{B}} = \overline{\mathbf{v}} \times \overline{\mathbf{B}} + \underbrace{\text{Cov}_W[\mathbf{v}, \mathbf{B}]_{\times}}_{\equiv \overline{\mathbf{v}' \times \mathbf{B}'}}. \quad (117)$$

We will interpret $\overline{\mathbf{v}' \times \mathbf{B}'}$ as a component of a *topology EMF* because it acts as an additional non-ideal source in the brane induction equation (see Section 8).

7.3 Mixed-sector forces and the physical role of A_w

The bulk Lorentz force for species s involves the full 4-spatial tensor F_{AB} with $A, B \in \{x, y, z, w\}$. Splitting the brane and transverse pieces gives (for brane components a):

$$q_s \left(E_a + F_{ab} v_s^b + F_{aw} v_s^w \right) = q_s \left(E_a + (\mathbf{v}_s \times \mathbf{B})_a + v_s^w C_a \right), \quad (118)$$

$$q_s \left(E_w + F_{wb} v_s^b \right) = q_s \left(E_w - v_s^a C_a \right), \quad (119)$$

using $C_a \equiv F_{aw}$ and antisymmetry of F_{AB} .

Equations (118)–(119) identify the new physical couplings beyond MHD:

- a **brane force channel** $q_s v_s^w C_a$ that vanishes identically when either $v_s^w = 0$ (no transverse drift) or $C_a = 0$ (no mixed structure);
- a **transverse acceleration channel** driven by E_w , which generically produces v_s^w and hence J^w .

Thus, suppressing A_w is not merely a gauge convenience: it removes a dynamical channel that can exchange energy and momentum between the brane-observed plasma and transverse degrees of freedom. In our simulation program, A_w is retained as dynamical except in controlled limit tests designed to reproduce textbook MHD benchmarks.

7.4 Finite localization and the transverse-mode tower

When $Z(w)$ is a Gaussian soft wall, the brane components A_μ admit a Hermite mode expansion in w (Appendix D), producing a tower of effective $3 + 1$ fields with masses $m_n^2 = 2n/\lambda^2$. The zero mode ($n = 0$) reproduces standard Maxwell. The higher modes ($n \geq 1$) yield:

1. **Yukawa corrections** to static interactions on scales $r \sim \lambda$;
2. **new storage/transport reservoirs** for EM energy and helicity, tracked mode-by-mode in the 3D×modes solver;
3. **a controlled truncation knob** N_w for massive computation: increasing N_w systematically increases transverse fidelity and should converge brane observables and leakage diagnostics.

A useful practical criterion is therefore the dimensionless ratio

$$\chi_\lambda \equiv \lambda |\nabla_3 \ln(\cdot)|, \quad (120)$$

evaluated on the relevant brane structures (current sheets, density gradients, etc.). When $\chi_\lambda \ll 1$, the zero-mode approximation is reliable and MHD-like behavior is expected. When $\chi_\lambda \gtrsim 1$, higher transverse modes and mixed-sector fields become dynamically relevant and beyond-MHD corrections should be expected.

7.5 Generalized Ohm’s law and the topology EMF

The brane induction equation is always obtained by projecting Faraday’s law:

$$\partial_t \bar{\mathbf{B}} = -\nabla_3 \times \bar{\mathbf{E}}. \quad (121)$$

Therefore, “non-ideality” on the brane is entirely controlled by how $\bar{\mathbf{E}}$ relates to the brane-facing flow variables and currents.

In standard two-fluid $\rightarrow$ MHD reductions, one obtains (schematically) $\mathbf{E} + \mathbf{v} \times \mathbf{B} = \eta \mathbf{J} + \dots$. In the present framework, two additional structures appear generically when transverse physics is active:

(i) Projection covariance EMF. Define the exact covariance EMF

$$\mathcal{E}_{\text{cov}} \equiv \overline{\mathbf{v} \times \mathbf{B}} - \bar{\mathbf{v}} \times \bar{\mathbf{B}} = \overline{\mathbf{v}' \times \mathbf{B}'}. \quad (122)$$

This term vanishes in the controlled MHD limit (no w -structure) but need not vanish even in laminar flows: it measures transverse variation, not turbulence.

(ii) Mixed-sector transverse EMF. From the Lorentz-force split (118), the electron equation contains the new term $v_e^w \mathbf{C}$. Its brane-facing contribution is the projected quantity

$$\mathcal{E}_w \equiv \overline{v_e^w \mathbf{C}}, \quad \mathbf{C} = (C_x, C_y, C_z), \quad C_a \equiv F_{aw}. \quad (123)$$

This term vanishes if either A_w is suppressed or transverse drift is absent; otherwise it provides a direct, computable mechanism for field-line slippage in brane observables.

Topology EMF. We group these contributions into a single topology-induced electromotive term

$$\mathcal{E}_{\text{topo}} \equiv \mathcal{E}_{\text{cov}} + \mathcal{E}_w + (\text{optional: additional two-fluid/closure pieces}). \quad (124)$$

Then a brane-facing induction equation can be written in the suggestive form

$$\partial_t \bar{\mathbf{B}} = \nabla_3 \times (\bar{\mathbf{v}} \times \bar{\mathbf{B}}) - \nabla_3 \times \mathcal{E}_{\text{topo}} + (\text{standard Hall/pressure/inertia/resistive terms if retained}). \quad (125)$$

Equation (125) is the operational statement of how the 4D interaction model extends MHD: topology change in brane observables is controlled by a diagnosable EMF sourced by transverse structure and mixed-sector dynamics. The associated energy and helicity ledgers, including explicit w -exchange and covariance reservoirs, are derived in Appendix E and form the basis for the validation tests in Section 10.

8 Generalized Ohm's Law with 4D/Projection Corrections

In standard two-fluid plasma theory, “non-ideality” on the brane is captured by a generalized Ohm law of the schematic form $\mathbf{E} + \mathbf{v} \times \mathbf{B} = \eta \mathbf{J} + \frac{\mathbf{J} \times \mathbf{B}}{en} - \frac{1}{en} \nabla_3 p_e + \dots$, and reconnection is associated with a nonzero $\mathbf{E} \cdot \mathbf{B}$ in localized regions. In the 4 + 1D interaction model, the same logic applies, but two new structures appear *even in the absence of ad hoc resistivity*:

1. **Mixed-sector (transverse) couplings** involving A_w , E_w , and $C_a \equiv F_{aw}$, which are absent in strictly 3 + 1 Maxwell theory.
2. **Projection-induced covariances** (“unresolved transverse structure”) because the brane observer measures $\overline{(\cdot)}$ rather than the pointwise bulk field.

The purpose of this section is to write these corrections explicitly in a form that can be computed in both the direct 4D solver and the 3D×modes solver.

8.1 Bulk electron momentum equation and mixed-sector decomposition

Let the electron species have charge $q_e = -e$ (with $e > 0$) and bulk number density $n_e(t, \mathbf{x}, w)$. Denote the bulk spatial velocity by

$$\mathbf{v}_e(t, \mathbf{x}, w) \equiv (v_e^x, v_e^y, v_e^z), \quad v_e^w(t, \mathbf{x}, w) \equiv \text{transverse component}. \quad (126)$$

Define the full 4D convective derivative

$$D_t^{(4)} \equiv \partial_t + v_e^a \partial_a + v_e^w \partial_w, \quad a \in \{x, y, z\}. \quad (127)$$

Using the bulk electromagnetic fields $E_a \equiv F_{a0}$, $B_a \equiv \frac{1}{2} \epsilon_{abc} F_{bc}$, and the mixed field $C_a \equiv F_{aw} = \partial_a A_w - \partial_w A_a$, the brane components of the bulk electron momentum equation may be written schematically as

$$m_e n_e D_t^{(4)} v_{e,a} = -en_e \left(E_a + (\mathbf{v}_e \times \mathbf{B})_a + v_e^w C_a \right) - \partial_a p_e + R_{e,a} + \dots, \quad (128)$$

where p_e is the electron pressure (or the corresponding enthalpy-gradient term in the GNLS realization), $R_{e,a}$ is the electron-ion momentum exchange term (optional), and $\dots$ denotes any additional modeled forces (e.g. dispersive/quantum terms if retained).

Solving (128) pointwise in w for $\mathbf{E}$ yields the *bulk* generalized Ohm law:

$$\mathbf{E} = -\mathbf{v}_e \times \mathbf{B} - v_e^w \mathbf{C} - \frac{1}{en_e} \nabla_3 p_e + \frac{1}{en_e} \mathbf{R}_e - \frac{m_e}{e} D_t^{(4)} \mathbf{v}_e + \dots, \quad (129)$$

where $\mathbf{C} \equiv (C_x, C_y, C_z)$ and $\mathbf{R}_e \equiv (R_{e,x}, R_{e,y}, R_{e,z})$.

New force channel (absent in 3 + 1). The term $-v_e^w \mathbf{C}$ is the direct imprint of the mixed field F_{aw} on the brane electric field. It vanishes in the controlled MHD limit (Section 6) but is generically present when A_w is dynamical and electrons drift in w .

8.2 Brane projection and exact projected Ohm form

Define the operational brane projection (Section 2) by

$$\overline{Q}(t, \mathbf{x}) \equiv \int_{-\infty}^{+\infty} dw W(w) Q(t, \mathbf{x}, w), \quad \int dw W(w) = 1. \quad (130)$$

Projecting (129) yields the *exact brane-facing identity*

$$\boxed{\overline{\mathbf{E}} = -\overline{\mathbf{v}}_e \times \overline{\mathbf{B}} - \overline{v_e^w} \overline{\mathbf{C}} - \frac{1}{en_e} \nabla_3 p_e + \frac{1}{en_e} \overline{\mathbf{R}}_e - \frac{m_e}{e} \overline{D_t^{(4)} \mathbf{v}_e} + \dots} \quad (131)$$

Equation (131) is not yet in the conventional MHD form because it contains projected products and projected ratios. This is not a defect: it is precisely the mathematical statement that the brane observer is not evolving a closed 3D system unless additional assumptions are imposed.

8.3 Quasi-neutral two-fluid rewrite and standard MHD terms

To connect (131) to the familiar two-fluid/MHD hierarchy, introduce brane-observed quasi-neutral variables

$$\overline{n} \equiv \overline{n_e} \simeq \overline{n_i}, \quad \overline{\mathbf{J}} \equiv e(\overline{n_i \mathbf{v}_i} - \overline{n_e \mathbf{v}_e}), \quad \overline{\mathbf{v}} \equiv (\text{single-fluid brane velocity}). \quad (132)$$

Under the usual ordering $m_e \ll m_i$, one has the standard approximation

$$\overline{\mathbf{v}} - \overline{\mathbf{v}}_e \approx \frac{\overline{\mathbf{J}}}{e \overline{n}}, \quad (133)$$

which produces the Hall term when multiplied by $\overline{\mathbf{B}}$.

Add and subtract $\overline{\mathbf{v}} \times \overline{\mathbf{B}}$ to obtain

$$\overline{\mathbf{E}} + \overline{\mathbf{v}} \times \overline{\mathbf{B}} = \frac{\overline{\mathbf{J}} \times \overline{\mathbf{B}}}{e \overline{n}} - \frac{1}{e \overline{n}} \nabla_3 \overline{p_e} + \eta \overline{\mathbf{J}} - \frac{m_e}{e} \mathcal{I}_e + \mathcal{E}_{\text{topo}} + \dots, \quad (134)$$

where:

- $\eta \overline{\mathbf{J}}$ is the usual resistive term if a drag model is used (Appendix C); otherwise set $\eta = 0$.
- $\mathcal{I}_e$ denotes the *chosen* electron-inertia closure in 3D variables (e.g. $\mathcal{I}_e \approx \overline{D_t^{(3)} \mathbf{v}_e}$ or an equivalent expression in $\overline{\mathbf{J}}$); its explicit form is not unique and depends on the closure target (ideal MHD, Hall MHD, extended MHD).
- $\mathcal{E}_{\text{topo}}$ collects the genuinely new 4D/projection terms, defined explicitly below.

Equation (134) is the primary “paper equation”: it shows that the brane system obeys a generalized Ohm law with the standard extended-MHD terms *plus* a new diagnosable electromotive contribution $\mathcal{E}_{\text{topo}}$.

8.4 Topology EMF and “topology stress” decomposition

Define the exact covariance operator (with respect to W) by

$$\text{Cov}_W[Q, R] \equiv \overline{QR} - \overline{Q}\overline{R}. \quad (135)$$

Then we define the topology EMF as

$$\boxed{\mathcal{E}_{\text{topo}} \equiv \mathcal{E}_{\text{cov}} + \mathcal{E}_w + \mathcal{E}_{\text{proj}}}, \quad (136)$$

with the three components:

$$\mathcal{E}_{\text{cov}} \equiv \overline{\mathbf{v}_e \times \mathbf{B}} - \overline{\mathbf{v}_e} \times \overline{\mathbf{B}} = -\text{Cov}_W[\mathbf{v}_e, \mathbf{B}]_{\times}, \quad (137)$$

$$\mathcal{E}_w \equiv -\overline{v_e^w \mathbf{C}}, \quad \mathbf{C} \equiv (F_{xw}, F_{yw}, F_{zw}), \quad (138)$$

$$\mathcal{E}_{\text{proj}} \equiv \left(-\frac{1}{en_e} \nabla_3 p_e + \frac{1}{e\bar{n}} \nabla_3 \bar{p}_e \right) + \left(\frac{1}{en_e} \mathbf{R}_e - \eta \bar{\mathbf{J}} \right) - \frac{m_e}{e} \left(\overline{D_t^{(4)} \mathbf{v}_e} - \mathcal{I}_e \right) + \dots. \quad (139)$$

Interpretation. $\mathcal{E}_{\text{topo}}$ is the part of $\overline{\mathbf{E}} + \overline{\mathbf{v}} \times \overline{\mathbf{B}}$ that is *forced* by transverse physics (mixed fields and w -structure) and by the fact that projection does not commute with nonlinear operations. It vanishes in the controlled MHD limit (Section 6), but it need not vanish even in laminar flows if there is transverse structure.

Why this acts like an “effective resistivity” without dissipation. In resistive MHD, non-ideal behavior is often modeled as $\mathbf{E} + \mathbf{v} \times \mathbf{B} \approx \eta \mathbf{J}$. Here, the term $\mathcal{E}_w = -\overline{v_e^w \mathbf{C}}$ has the same structural role in the induction equation (it contributes through a curl), but it corresponds to *momentum/energy export into transverse degrees of freedom* rather than collisional heating. This distinction is made quantitative by the energy and helicity ledgers with explicit leakage (Appendix E).

8.5 Induction equation and reconnection diagnostics

Projecting Faraday’s law gives

$$\partial_t \overline{\mathbf{B}} = -\nabla_3 \times \overline{\mathbf{E}}. \quad (140)$$

Using (134), the brane induction equation becomes

$$\boxed{\partial_t \overline{\mathbf{B}} = \nabla_3 \times (\overline{\mathbf{v}} \times \overline{\mathbf{B}}) - \nabla_3 \times \left[\frac{\overline{\mathbf{J}} \times \overline{\mathbf{B}}}{e\bar{n}} - \frac{1}{e\bar{n}} \nabla_3 \bar{p}_e + \eta \bar{\mathbf{J}} - \frac{m_e}{e} \mathcal{I}_e + \mathcal{E}_{\text{topo}} \right] + \dots}, \quad (141)$$

with $\nabla_3 \cdot \overline{\mathbf{B}} = 0$ maintained by the Maxwell solver (or by the chosen divergence-control scheme).

A key reconnection/topology diagnostic is the component of the non-ideal electric field parallel to the magnetic field,

$$\overline{E}_{\parallel} \equiv \frac{\overline{\mathbf{E}} \cdot \overline{\mathbf{B}}}{|\overline{\mathbf{B}}|}, \quad \overline{E}_{\parallel} = \left[\eta \bar{\mathbf{J}} - \frac{m_e}{e} \mathcal{I}_e + \mathcal{E}_{\text{topo}} + \dots \right] \cdot \frac{\overline{\mathbf{B}}}{|\overline{\mathbf{B}}|}, \quad (142)$$

showing explicitly that a nonzero $\overline{E}_{\parallel}$ can arise from the topology EMF even when $\eta = 0$. The associated energy/helicity accounting is handled by the projected budgets in Appendix E.

8.6 How $\mathcal{E}_{\text{topo}}$ is computed in simulations

The decomposition (136) is designed to be *computationally explicit*:

- **Direct 4D-grid solver.** All quantities in (137)–(139) are evaluated on the (x, y, z, w) grid and then integrated against $W(w)$ to form $(\cdot)$.
- **3D×modes solver (Hermite / soft-wall).** Reconstruct the needed fields at transverse quadrature nodes w_q from the retained mode amplitudes (Appendix D), form the nonlinear products $\mathbf{v}_e \times \mathbf{B}$ and $v_e^w \mathbf{C}$ pointwise in w_q , and then apply quadrature for $(\cdot)$ and for the covariance differences. Mode-truncation convergence is assessed by increasing N_w until $\mathcal{E}_{\text{topo}}$ and the reconnection/helicity diagnostics stabilize (Section 10).

In both architectures, $\mathcal{E}_{\text{topo}}$ is therefore not an *assumed closure* but a directly measured quantity arising from 4D interaction structure and operational projection.

9 Computation: How We Will Do Massive Plasma Simulations

This section translates the model hierarchy into a concrete numerical program. The primary objective is to enable *massive* plasma simulations that retain the essential 4D interaction channels (transverse transport in w , scalar-photon dynamics A_w , and transverse-mode excitation) while remaining computationally tractable on modern HPC architectures. We therefore present two solver families:

1. **Direct 4D-grid solvers** that discretize (x, y, z, w) explicitly. These are the most direct representation of the bulk PDEs and serve as “ground truth” and validation for reduced models.
2. **3D×modes solvers** that discretize (x, y, z) on a large grid and represent w -dependence in a truncated spectral basis (Hermite modes for Gaussian localization). This is the recommended approach for truly massive runs.

Throughout, we avoid evolving only brane variables: the computational strategy is to evolve bulk degrees of freedom and then compute brane observables by the projection map $\bar{\cdot}$, including explicit leakage diagnostics.

9.1 State variables and discretization choices

Electromagnetic variables

We evolve the gauge potential $A_M(t, \mathbf{x}, w)$ with a hyperbolic gauge choice (e.g. generalized Lorenz gauge as in (71)) so that the field equations form a well-posed time-domain system. Evolving A_M has two practical advantages:

1. the homogeneous Maxwell equations (Bianchi identity) are satisfied by construction;
2. the mixed sector $(A_w, F_{\mu w})$ is handled on the same footing as the brane sector without additional constraint equations.

An alternative (useful for some plasma codes) is to evolve field strengths (E_A, F_{AB}) with explicit constraint cleaning. In this paper we prioritize the potential formulation because the transverse-mode decomposition (Sec. D) is naturally expressed in terms of A_M .

Plasma variables

We support three plasma representations, ordered by increasing kinetic fidelity:

(i) **Two-fluid in 4D space.** Evolve $(n_s, \mathbf{v}_s)$ on the bulk spatial domain $(\mathbf{x}, w)$, with $\mathbf{v}_s \in \mathbb{R}^4$, and close pressure by a specified EOS. This is the closest analogue to extended MHD, but in 4 spatial dimensions.

(ii) **Field-based (multi-species GNLS).** Evolve $\psi_s(t, \mathbf{x}, w)$ for each species and compute $\rho_s = |\psi_s|^2$ and J_s^M from (75)–(76). This approach evolves canonical momentum naturally (Sec. 4.3) and remains well-defined at low density, which is attractive for topology change.

(iii) **Kinetic (4D Vlasov / PIC).** Evolve particle distributions in 4D space with 4D velocity. This is the most expensive option but provides the clearest path to collisionless reconnection physics in regimes where fluid closures are insufficient.

Spatial discretization in (x, y, z)

For brane spatial directions we assume a large uniform grid for simplicity and HPC efficiency, though adaptive mesh refinement is compatible with the mode-based approach. We recommend:

- **finite differences** (2nd–4th order) for Maxwell in potentials;
- **finite volume** (Godunov-type) for two-fluid bulk equations when shocks or steep gradients appear;
- **spectral/FFT Poisson solves** only in regimes where a reduced Maxwell closure is used (e.g. Darwin/MHD ordering), not in the full wave evolution.

Representation of the extra dimension w

Two choices:

Direct w -grid (baseline). Use a finite interval $w \in [-W_{\max}, W_{\max}]$ with boundary conditions chosen to respect localization (e.g. outgoing/radiative boundaries). This is simple but can be expensive because stable timesteps are constrained by the finest Δw .

Hermite spectral in w (recommended). When $Z(w)$ is Gaussian, the transverse operator admits a Hermite basis with a diagonal mass spectrum $m_n^2 \propto n$ (Sec. D). A truncated mode expansion converts the problem into a set of coupled $3 + 1$ PDEs for mode amplitudes. This is the primary pathway to massive simulations.

9.2 Hermite spectral decomposition in w

Field expansion and truncation

For each component of the gauge potential (and, when desired, for matter fields), we use a truncated expansion

$$A_M(t, \mathbf{x}, w) \approx \sum_{n=0}^{N_w-1} a_M^{(n)}(t, \mathbf{x}) \phi_n(w), \quad (143)$$

where $\{\phi_n(w)\}$ are orthonormal with respect to the Z -weighted inner product $\langle f, g \rangle_Z = \int dw Z(w) f(w) g(w)$. For Gaussian Z , a Hermite basis yields a simple spectrum

$$m_n^2 = \frac{2n}{\lambda^2}, \quad n = 0, 1, 2, \dots, \quad (144)$$

and brane coupling selection rules (odd modes decouple from strictly brane-localized sources).

The truncation N_w is a *physical* knob: increasing N_w increases the available transverse degrees of freedom for topology and energy to flow into/out of the brane-observed sector. Convergence is assessed by monitoring the mode-energy spectrum and leakage diagnostics (Sec. 9.5).

Linear operators are diagonal in mode index

Under standard gauge conditions (e.g. Lorenz gauge), the linear Maxwell operator reduces mode-by-mode to brane wave operators of the form

$$(\Box_4 + m_n^2) a_\mu^{(n)}(t, \mathbf{x}) = (\text{mode-projected sources}), \quad \Box_4 \equiv -\partial_t^2 + \nabla_3^2. \quad (145)$$

Thus the expensive part of the solver (the linear field propagation) becomes a collection of independent, stencil-based 3D wave updates indexed by n , which is ideal for vectorization, GPUs, and mode-parallel MPI.

Nonlinear couplings handled pseudo-spectrally

Nonlinearities enter through sources $J^M(\psi_s, A)$ and/or through fluid nonlinear terms. Rather than deriving dense mode-coupling tensors in closed form, we use a pseudo-spectral strategy in w :

1. choose quadrature points $\{w_q\}_{q=1}^{Q_w}$ and weights $\{\omega_q\}$ appropriate to the Z -weighted inner product;
2. reconstruct fields at quadrature nodes: $A_M(t, \mathbf{x}, w_q) = \sum_n a_M^{(n)}(t, \mathbf{x}) \phi_n(w_q)$ and similarly for matter variables;
3. compute nonlinear source terms pointwise at each w_q ;
4. project back to modes by quadrature:

$$J^{M,(n)}(t, \mathbf{x}) \approx \sum_{q=1}^{Q_w} \omega_q Z(w_q) \phi_n(w_q) J^M(t, \mathbf{x}, w_q). \quad (146)$$

This keeps the mode truncation controllable and avoids introducing spurious “hard-wall” artifacts in w .

9.3 Coupling algorithm (field solve + plasma push)

We use a standard split-step / operator-splitting approach tailored to the chosen plasma model. A typical timestep from $t^n \rightarrow t^{n+1} = t^n + \Delta t$ proceeds as:

1. **Reconstruct fields in w (mode solver only):** build $A_M(t^n, \mathbf{x}, w_q)$ at quadrature nodes w_q if using the 3D $\times$ modes scheme.
2. **Advance plasma (matter step):**
 - *Two-fluid*: update $(n_s, \mathbf{v}_s)$ with a conservative finite-volume scheme in (x, y, z) for each w_q or mode coefficient, including Lorentz force terms using F_{MN} .
 - *GNLS*: apply a Strang split: kinetic covariant step $-(\hbar^2/2m_s)D_A D_A$, then local non-linear/potential step $(V_{\text{conf},s} + h_s)$, with gauge fields held fixed within substeps.
 - *PIC/Vlasov*: push particles in 4D space using the Lorentz tensor, deposit charge/current with a charge-conserving scheme to guarantee discrete $\partial_M J^M = 0$.

3. **Compute sources:** evaluate J^M from updated matter variables. In mode form this yields $\{J^{M,(n)}(t^{n+\frac{1}{2}}, \mathbf{x})\}$ by projection (pseudo-spectral step).
4. **Advance Maxwell (field step):** update the localized Maxwell system in potentials. In 3D $\times$ modes form, each mode is updated with a wave-like stencil including the mass term m_n^2 . In the direct 4D-grid form, update $A_M(t, \mathbf{x}, w)$ with the variable-coefficient operator $\partial_M(ZF^{MN})$.
5. **Constraint/gauge control:** apply gauge-driver or constraint-damping steps (e.g. generalized Lorenz gauge damping) to suppress numerical drift in $\partial \cdot A$. If evolving fields directly, apply divergence cleaning to maintain Gauss constraints.
6. **Diagnostics and brane observables:** compute brane-projected fields $\overline{F_{\mu\nu}}$, brane currents, and leakage terms $S_{\text{leak}}^{(s)}$. Record energy budgets and mode spectra (Sec. 9.5).

Time stepping and stiffness. Full Maxwell evolution carries the light-wave CFL condition. For MHD-like regimes where fast light waves are not of primary interest, one may employ IMEX time stepping (implicit Maxwell, explicit plasma) or reduced-field closures (Darwin / low-frequency approximations) while retaining the transverse-mode structure in w . The present paper focuses on the full hyperbolic system and treats reduced closures as optional accelerations once correctness is established.

9.4 Parallelization and scaling considerations

Cost model

Let $N_x N_y N_z$ be the brane grid size, and let N_w be either the number of w -grid points (direct 4D) or the number of transverse modes (3D $\times$ modes). Then, schematically:

- **Direct 4D grid:** memory and compute scale as $\mathcal{O}(N_x N_y N_z N_w)$, with Maxwell CFL constrained by $\min(\Delta x, \Delta w)$.
- **3D $\times$ modes:** memory and compute scale as $\mathcal{O}(N_x N_y N_z N_w)$ as well, but with substantially smaller prefactors because: (i) the w -operator is diagonal in mode index for the linear step, and (ii) the reconstruction/projection in w is a dense operation only over a small N_w (tens), not over a large w -grid (hundreds or thousands).

In practice, massive runs target $N_w \sim 8$ to 64 modes, chosen by convergence of the mode-energy spectrum and leakage diagnostics.

Domain decomposition and mode parallelism

We use two independent decomposition axes:

1. **Spatial decomposition in (x, y, z) :** standard MPI halo exchange for stencil updates of Maxwell and fluid variables.
2. **Mode decomposition in n :** distribute subsets of mode indices across ranks (or GPU streams). The linear Maxwell update is embarrassingly parallel in n . Nonlinear reconstruction/projection requires only local (per-cell) operations and therefore minimal communication.

A hybrid layout (space decomposition outer, mode decomposition inner) enables both strong scaling (more ranks for fixed problem) and weak scaling (larger grids with fixed per-rank memory).

GPU suitability

The $3D \times \text{modes}$ representation is particularly GPU-friendly because each cell carries short vectors of mode coefficients $\{a_M^{(n)}\}_{n=0}^{N_w-1}$ and the dominant kernels are fused per-cell updates:

- mode-local Maxwell stencils (vector length N_w),
- reconstruction/projection in w (small dense matrix multiply per cell),
- current construction for each species.

9.5 Diagnostics for reconnection and topology transport

A central claim of this framework is that brane-observed “non-ideality” can arise from conservative transport into transverse degrees of freedom rather than from ad hoc resistivity. Simulations must therefore record diagnostics that separate: (i) true dissipation (if collisions are included) from (ii) topology/energy export into w and into higher transverse modes.

Bulk and mode-resolved energy budgets

We track:

- **Total bulk energy** $\mathcal{E}_{\text{tot}}(t)$ from (82) as a global correctness check.
- **Mode energies** $E_{\text{EM}}^{(n)}(t)$ in the $3D \times \text{modes}$ solver, computed from the quadratic forms induced by the action (including mass terms m_n^2). A growth of $n \geq 1$ energy quantifies transverse-mode excitation.
- **Scalar-photon channel energy** associated with A_w and $E_w = F_{w0}$, including the work term $J^w E_w$ in the exchange identity (83).

Leakage (open-system) diagnostics

For each species s , we compute the projected leakage source $S_{\text{leak}}^{(s)}$ from (22). In addition we record bulk transverse fluxes:

$$\Phi_s^{(w)}(t; \mathbf{x}) \equiv j_s^w(t, \mathbf{x}, w), \quad \overline{\Phi_s^{(w)}}(t, \mathbf{x}) = \int dw W(w) j_s^w(t, \mathbf{x}, w), \quad (147)$$

as well as the integrated net exchange over a region $\Omega \subset \mathbb{R}^3$:

$$\mathcal{L}_s(t; \Omega) \equiv \int_{\Omega} S_{\text{leak}}^{(s)}(t, \mathbf{x}) d^3x. \quad (148)$$

Nonzero $\mathcal{L}_s$ directly measures brane–bulk exchange at the operational measurement scale.

Reconnection-rate proxies (brane-facing)

On the brane we compute standard reconnection indicators from brane-projected fields (e.g. $\mathbf{E}^{(\text{brane})}$, $\mathbf{B}^{(\text{brane})}$) while interpreting their non-ideal parts through bulk channels:

- **Parallel electric field proxy:** $E_{\parallel}^{(\text{brane})} \equiv \mathbf{E}^{(\text{brane})} \cdot \mathbf{B}^{(\text{brane})} / |\mathbf{B}^{(\text{brane})}|$, and its line integral through current sheets.
- **Brane helicity change:** compute magnetic helicity $H_{\text{brane}} = \int d^3x \mathbf{A}^{(\text{brane})} \cdot \mathbf{B}^{(\text{brane})}$ in a chosen brane gauge and track dH_{brane}/dt . Changes in H_{brane} are compared against energy transfer into A_w , E_w , and higher modes to distinguish topology export from dissipation.
- **Topology-stress signature:** evaluate projected unresolved-stress terms (e.g. the “topology stress tensor” defined later in Section 8) to quantify how sub- w structure drives effective non-ideal terms in brane balances.

Convergence criteria specific to this framework. A simulation is regarded as “4D-converged” not only when brane-facing fields are grid-converged in (x, y, z) , but also when: (i) leakage diagnostics stabilize under increases in N_w , (ii) mode-energy spectra show a resolved cascade or decay with n , and (iii) the bulk energy ledger closes to within the expected numerical tolerance. These criteria are essential because reconnection-like behavior may manifest as transport into higher transverse modes rather than as local heating.

10 Validation Plan and Benchmark Problems

This section outlines a validation program designed to establish correctness at three distinct levels:

1. **Bulk PDE correctness:** the $4 + 1$ Maxwell–matter system is solved accurately, respects discrete charge conservation, satisfies the homogeneous Maxwell identities, and closes the bulk energy ledger.
2. **Controlled-limit correctness:** when the controlled assumptions in Section 6 are enforced (zero-mode dominance, negligible $F_{\mu w}$ and J^w , quasi-neutral two-fluid ordering), the simulation reduces to standard 3D Maxwell/two-fluid/MHD benchmark behavior.
3. **Beyond-MHD signal isolation:** when 4D interaction channels are activated (A_w , $F_{\mu w}$, J^w , transverse-mode tower), their effects appear in the predicted observables (leakage sources, mode-energy transfer, topology-stress signatures) while the bulk system remains conservative.

We explicitly validate both computational architectures described in Section 9: **direct 4D-grid** evolution in (x, y, z, w) and the **3D×modes** Hermite spectral reduction in w . A major goal is to show that the mode solver converges to the direct solver as N_w is increased.

10.1 MHD benchmarks (controlled limit)

All tests in this subsection are run in a controlled-limit configuration:

$$\partial_w(\cdot) \approx 0, \quad A_w \approx 0, \quad F_{\mu w} \approx 0, \quad J^w \approx 0, \quad N_w = 1 \text{ (mode solver)}. \quad (149)$$

In this configuration, brane projection becomes trivial ($\bar{Q} \approx Q$) and the effective coupling $\mu_0 \rightarrow \mu_0^{\text{eff}}$ is fixed by Z_{int} as in (89).

(A) Maxwell-sector verification in $3 + 1$

1. **Vacuum plane waves.** Initialize a vacuum wave packet in Lorenz gauge and verify: (i) wave speed c (or 1 in units), (ii) preservation of the gauge condition, (iii) convergence rates under grid refinement, and (iv) closure of electromagnetic energy to truncation error.
2. **Static Coulomb field.** Place a stationary brane charge distribution $\rho_q(\mathbf{x})$ and verify the static solution $\mathbf{E} = -\nabla_3\phi$ reproduces the Poisson limit of (92) (with ϵ_0^{eff}). This test is useful for verifying charge deposition, Gauss constraints, and solver correctness.
3. **Charge conservation under a moving source.** Evolve a prescribed moving brane charge/current that is analytically conserved and verify discrete $\partial_t \rho_q + \nabla_3 \cdot \mathbf{J} = 0$ and stability of Gauss constraints.

(B) Two-fluid and ideal-MHD wave benchmarks

1. **Circularly polarized Alfvén wave.** Initialize a nonlinear Alfvén wave (a standard code verification target) and verify amplitude preservation, phase speed, and convergence.
2. **Fast/slow magnetosonic waves.** Initialize linear perturbations about a uniform background and verify dispersion relations in the two-fluid solver (and in the ideal-MHD closure when invoked).
3. **Whistler/Hall branch (optional extended-MHD check).** If the generalized Ohm law (99) is used with the Hall term retained, verify the whistler dispersion relation. This provides a sensitive check of $\mathbf{J} \times \mathbf{B}$ coupling and current evaluation.

(C) Standard 3D MHD code benchmarks

These tests validate shock capturing, nonlinear coupling, and topology handling in the controlled limit:

1. **Brio–Wu shock tube (1D MHD Riemann problem).** Verify shock/rarefaction structures and convergence to a reference solution.
2. **Orszag–Tang vortex (2D).** A standard benchmark for transition to MHD turbulence. Validate global invariants, magnetic/kinetic energy evolution, and compare with reference MHD results.
3. **Field-loop advection (2D/3D).** Advect a weak magnetic loop in a prescribed velocity field and verify minimal numerical diffusion and preservation of $\nabla_3 \cdot \mathbf{B} = 0$.
4. **Kelvin–Helmholtz instability (2D/3D).** Validate linear growth rates (early time) and nonlinear roll-up (late time), with attention to divergence control and energy conservation.

(D) Controlled reconnection tests (resistive and/or Hall MHD)

To compare with known reconnection scalings in the controlled limit:

1. **Harris current sheet (2D).** Run tearing-mode onset with either explicit resistivity η or Hall terms. Measure reconnection rate versus η (Sweet–Parker scaling) and/or Hall length.
2. **GEM reconnection challenge configuration (optional).** If kinetic or two-fluid physics is active, compare reconnected flux versus time and outflow speeds to the canonical GEM setup as a reference behavior target.

Acceptance criteria for controlled-limit benchmarks. A controlled-limit validation pass requires:

1. correct linear dispersion (waves) within resolution error;
2. correct shock structure (Riemann problems) and convergence with refinement;
3. $\|\nabla_3 \cdot \mathbf{B}\|$ bounded and decreasing with refinement (or maintained to machine tolerance with constrained transport);
4. agreement with standard MHD reference outputs to within discretization error.

10.2 Beyond-MHD tests (4D effects)

The objective here is not to reproduce standard MHD results, but to validate that the *new* degrees of freedom behave as predicted, and that their brane-facing signatures can be isolated cleanly while the bulk system remains conservative. Unless otherwise stated, these tests are performed in the full $4 + 1$ formulation (direct w -grid or $3D \times \text{modes}$ with $N_w > 1$) and brane observables are computed via $\bar{\cdot}$.

(A) Scalar-photon channel: free evolution and coupling

1. **Vacuum A_w pulse propagation.** Initialize $A_w(t = 0, \mathbf{x}, w)$ as a localized pulse (with all sources off) and verify: (i) stable propagation under the localized Maxwell operator, (ii) correct energy partition in the mixed sector via $E_w = F_{w0}$ and $C_a = F_{aw}$, and (iii) gauge-driver stability (no spurious growth of $\partial \cdot A$).
2. **Matter response to E_w .** Initialize a species density localized near the brane and apply a controlled A_w configuration that produces a nonzero E_w . Verify that a transverse drift $v_{s,w}$ develops in accordance with the bulk momentum equation (68), and that the resulting j_s^w generates a measurable brane leakage $S_{\text{leak}}^{(s)}$ through (22).

(B) Leakage identity tests (open-system brane balance laws)

These tests validate that the brane is an open subsystem and that the leakage terms are quantitatively correct:

1. **Continuity leakage closure.** Construct an initial condition with a known, imposed $j_s^w(t, \mathbf{x}, w)$ (e.g. a transverse pulse) and verify that the measured brane density obeys

$$\partial_t \rho_{s,\text{brane}} + \partial_a j_{s,\text{brane}}^a = S_{\text{leak}}^{(s)} \quad (150)$$

to numerical accuracy. This is an exact identity test (discrete verification of (21)).

2. **Bulk conservation with nonzero brane leakage.** Simultaneously verify that the bulk continuity equation remains satisfied while the brane subsystem exhibits net exchange $\mathcal{L}_s(t; \Omega) \neq 0$ for an appropriate brane region Ω . This demonstrates that brane non-ideality is consistent with conservative bulk evolution.

(C) Finite-localization and mode-tower tests

These tests validate the Hermite/KK tower structure and its physical consequences:

1. **Static point-source potential (Coulomb + Yukawa tower).** For a brane-localized static charge, compute the brane potential $A_0(r)$ and verify agreement with the predicted Coulomb-plus-Yukawa form (e.g. (222)) in the regimes $r \gg \lambda$ and $r \sim \lambda$. This directly validates mode masses m_n and coupling weights c_n .
2. **Mode excitation spectrum under sharp gradients.** Drive the system with a brane-localized current sheet whose thickness is varied across $\sim \lambda$. Verify that: (i) the $n = 0$ mode dominates for broad structures, (ii) higher modes are excited as gradients approach λ^{-1} , and (iii) the mode-energy spectrum converges as N_w increases.

(D) Reconnection via topology export (key beyond-MHD demonstration)

The flagship beyond-MHD validation is to show that reconnection-like topology change in brane observables can occur through conservative 4D transport channels:

1. **Current-sheet reconnection with suppressed resistivity.** Initialize a brane current sheet in a configuration that would require explicit resistivity (or numerical diffusion) to reconnect in ideal MHD. Run two variants:
 - *Controlled-limit run:* enforce $A_w \approx 0$, $J^w \approx 0$, $N_w = 1$. Reconnection should be suppressed up to numerical diffusion.
 - *Full 4D-interaction run:* allow A_w , $F_{\mu w}$, J^w , and/or $N_w > 1$. Measure enhanced reconnection signatures on the brane.

The validation target is not a single reconnection rate, but a *ledger closure*: the increase in brane reconnected flux must correlate with: (i) nonzero transverse exchange (S_{leak} , $J^w E_w$), (ii) growth of mixed-sector energy (via E_w , F_{aw}), and/or (iii) transfer of EM energy into higher transverse modes.

2. **Topology-stress correlation.** Compute the projected unresolved-stress / “topology stress tensor” terms defined in Section 8 and verify that their support localizes near reconnection regions and correlates with the non-ideal term in the brane induction equation inferred from $\partial_t \mathbf{B} + \nabla_3 \times (\mathbf{v} \times \mathbf{B})$.

Acceptance criteria for beyond-MHD tests. A beyond-MHD validation pass requires:

1. exact projected identities (e.g. leakage balance) hold discretely to solver tolerance;
2. bulk global energy $\mathcal{E}_{\text{tot}}(t)$ remains conserved up to expected numerical error when collisions are off;
3. mode spectra and leakage diagnostics converge as N_w and Q_w increase;
4. reconnection-like brane behavior is accompanied by quantifiable transport into w /modes (not solely by numerical diffusion).

10.3 Convergence and truncation studies

To support massive simulations, we require convergence studies across four axes:

1. **Brane spatial resolution:** refine Δx in (x, y, z) and verify expected convergence rates for smooth problems and correct shock capture for discontinuous problems.
2. **Time step:** refine Δt and verify stability and convergence, including sensitivity to gauge-driver parameters.
3. **Transverse resolution:** increase either W_{max} and Δw (direct solver) or N_w and quadrature order Q_w (mode solver) until brane observables and diagnostics converge.
4. **Solver consistency:** demonstrate that the 3D $\times$ modes solver converges to the direct 4D-grid solver for shared test problems as N_w is increased and/or as the direct w -grid is refined.

Recommended convergence reporting. For each benchmark we report: (i) $\|\nabla_3 \cdot \mathbf{B}\|$ norms, (ii) bulk energy closure error, (iii) brane leakage norm $\|S_{\text{leak}}\|$ and integrated exchange $\mathcal{L}_s$, (iv) mode-energy spectra $E_{\text{EM}}^{(n)}$, and (v) brane reconnection proxies (when applicable). This provides a unified basis for distinguishing true physical beyond-MHD behavior from truncation artifacts.

11 Discussion

This paper advances a “4D-first” viewpoint for plasma modeling: the fundamental dynamics lives in a conservative bulk with four spatial dimensions (x, y, z, w) , and the familiar 3D plasma variables arise as *brane-facing observables* defined by a projection map $\bar{\cdot}$. In this setting, standard MHD is not replaced; it is recovered as a controlled limit. The primary payoff is that the mechanisms normally inserted into MHD as phenomenological non-idealities acquire a geometric and dynamical origin: topology change on the brane can correspond to conservative transport into transverse degrees of freedom in w and into the transverse mode tower induced by localization.

11.1 MHD as a controlled brane/zero-mode limit

Section 6 makes explicit that recovering MHD requires more than taking a low-frequency limit. One must additionally suppress the *degrees of freedom that are absent in strictly 3 + 1 electromagnetism*:

$$\partial_w(\cdot) \approx 0, \quad F_{\mu w} \approx 0, \quad J^w \approx 0, \quad (151)$$

together with the standard two-fluid $\rightarrow$ MHD assumptions (quasi-neutrality and electron inertia suppression). In this regime, the localized Maxwell sector reduces to a 3 + 1 Maxwell system with effective coupling $\mu_0^{\text{eff}} = \mu_0 / \int Z dw$, and the usual ideal induction equation follows. The controlled-limit derivation is important because it cleanly identifies what must be “turned back on” to obtain new plasma physics: transverse current J^w , mixed field components $F_{\mu w}$, and excitation of higher transverse modes.

11.2 Beyond MHD as conservative topology and energy transport

A key conceptual distinction of this framework is the difference between:

1. **Dissipation to heat** (collisional resistivity, viscosity, Landau damping), which changes entropy and reduces macroscopic free energy; and
2. **Export into additional degrees of freedom**, which can appear as “effective non-ideality” in brane-facing equations while leaving the bulk system conservative.

The latter is the characteristic novelty of the 4D interaction model. When A_w , E_w , F_{aw} , and transverse modes are dynamically active, the brane subsystem generically becomes an *open* subsystem with explicit exchange terms. From the brane perspective, those exchange terms can mimic the role usually played by resistivity in enabling field-line slippage and reconnection, but without requiring local conversion of magnetic energy directly into heat. Instead, energy may be routed into mixed-sector fields and/or into higher transverse modes, and later returned.

This interpretation suggests a physically sharp diagnostic criterion for distinguishing “numerical reconnection” from “4D-enabled reconnection” in simulations: reconnection-like brane signatures should correlate with measurable transport into transverse channels, such as nonzero work $J^w E_w$, growth of mixed-sector energy, and/or growth of mode energies with $n \geq 1$ in the Hermite/KK decomposition.

11.3 Relationship to extended MHD and kinetic plasma theory

On the brane, the standard extended-MHD hierarchy appears in the usual way: Hall drift, electron pressure, electron inertia, and collisional resistivity arise from controlled relaxations of the two-fluid assumptions in Section 6.2. In this sense, the present model is compatible with (and can reproduce) standard extended-MHD closures.

What is new is that additional correction terms arise even when conventional extended-MHD closures are held fixed, simply because the parent theory contains more degrees of freedom than $3 + 1$ Maxwell:

- the scalar-photon channel A_w and its associated fields E_w and F_{aw} ,
- explicit brane–bulk exchange via J^w (leakage sources in projected balances),
- and finite-localization mode-tower corrections (massive modes with $m_n^2 \propto n$).

In the language of generalized Ohm’s law, these effects appear as additional terms beyond Hall/pressure/inertia, including mixed-field forces and projected unresolved stresses (Section 8). This suggests a unifying view: *extended MHD* expands the matter closure at fixed $3 + 1$ EM, while the *4D interaction model* expands the EM and topology sector itself.

The framework also provides a natural bridge to kinetic methods. In the kinetic (Vlasov/PIC) realization, the “massive simulation” strategy becomes: evolve the full 3D phase-space dynamics while representing transverse structure in w by a truncated mode basis (or a modest direct w -grid), thereby retaining the new topology channels without requiring an intractable full 4D spatial grid at high resolution.

11.4 Canonical vorticity constraint and the origin of brane non-ideality

A useful structural identity in the bulk is the London-type relation (69):

$$\partial_A v_{s,B} - \partial_B v_{s,A} = -\frac{q_s}{m_s} F_{AB}, \quad (A, B \in \{x, y, z, w\}), \quad (152)$$

valid away from phase singularities in the field-based realization (and interpreted as a canonical-momentum constraint more generally). This identity can be read as a geometric “flux locking” statement: canonical vorticity is tied to gauge flux in the bulk. A brane observer, however, does not measure the full bulk tensor F_{AB} ; it measures projected quantities $\overline{F}_{\mu\nu}$ and experiences explicit exchange terms when $J^w \neq 0$. Thus, what appears as a violation of ideal constraints on the brane can correspond to perfectly consistent bulk evolution plus projection. This is the mathematical origin of the “topology export” interpretation of reconnection in this framework.

11.5 The meaning of w , Z , and W

The model separates two roles often conflated in dimensional reduction:

- $Z(w)$ controls how strongly electromagnetic dynamics is localized near the operational brane. It sets an intrinsic transverse length scale λ (for Gaussian localization) and thereby controls when the mode tower becomes important.
- $W(w)$ defines what the brane observer measures. It can be matched to $Z/\int Z$ for a natural “measure where the field lives” choice, or chosen differently to represent finite measurement resolution or a particular observational protocol.

In simulation practice, λ and the mode truncation N_w are therefore not merely numerical parameters; they are physical modeling knobs that control the strength of beyond-MHD corrections. A major objective for future work is to calibrate these knobs against known plasma scales (e.g. current-sheet thickness, electron skin depth, or other microphysical lengths) in specific regimes, thereby making the 4D interaction channels quantitatively predictive.

11.6 Falsifiable signatures and near-term simulation targets

The framework makes several concrete, testable predictions at the level of simulation diagnostics:

1. **Mode-spectrum signature:** reconnection-like brane behavior should correlate with energy transfer from the $n = 0$ Maxwell mode into higher transverse modes $n \geq 1$ when structures approach the localization scale $r \sim \lambda$. In the Gaussian case, the spectrum and coupling selection rules are fixed (odd modes decouple from strictly brane-localized sources), providing a sharp test of correct transverse modeling.
2. **Scalar-photon signature:** activation of the A_w sector should be detectable as nontrivial mixed-field energy (via E_w and F_{aw}) and nonzero work channel $J^w E_w$, particularly near strong-gradient regions.
3. **Leakage-topology correlation:** changes in brane helicity (or other topological diagnostics) should correlate with explicit brane exchange measures such as $\|S_{\text{leak}}\|$ and integrated leakage $\mathcal{L}_s(t; \Omega)$, rather than correlating solely with collisional heating.
4. **Topology-stress localization:** the projected unresolved-stress contribution (the “topology stress tensor” defined in Section 8) should localize near reconnection regions and provide a quantitative accounting of the non-ideal term in the brane induction equation.

These signatures are well suited to the validation/benchmark program in Section 10 and provide a clear path to assessing whether 4D interaction effects are being captured rather than replaced by numerical diffusion.

11.7 Limitations and future work

The present work has several limitations that define a roadmap:

- **Matter-sector regime.** The baseline matter model is nonrelativistic. This is appropriate for MHD-like regimes but will require upgrade (relativistic kinetic matter) for ultra-relativistic plasmas or extreme-field environments.
- **Closure choices.** The two-fluid reduction uses standard closures (scalar pressures, optional collisional drag). Incorporating anisotropic pressure tensors, heat flux closures, and collision operators consistent with the bulk conservation laws is a direct extension.
- **Transverse calibration.** While the transverse spectral structure is fixed once $Z(w)$ is chosen, mapping λ and N_w to physical plasma scales in specific regimes remains an open calibration problem and a central target for future numerical experiments.
- **Numerical realizations.** The 3D×modes algorithm is designed for massive simulations, but its robustness across strongly nonlinear regimes must be established by systematic convergence studies and comparison to direct 4D-grid reference runs.
- **Geometry and additional sectors.** The present paper treats $Z(w)$ as fixed. Allowing back-reaction of plasma loads on localization/geometry degrees of freedom (when present in the broader unified model) is a natural next step, as is coupling to gravitational sectors when appropriate.

Despite these limitations, the model provides a coherent hierarchy: a conservative bulk theory, an operational brane map, a controlled MHD limit, and a computable set of beyond-MHD correction channels with clear simulation knobs. This hierarchy is designed to make reconnection and other topology-changing plasma processes not a special exception requiring ad hoc terms, but an emergent, diagnosable consequence of evolving the full 4D interaction structure.

12 Conclusion

We have presented a practical “4D-first” framework for plasma physics in which the fundamental interaction dynamics is evolved in a conservative 4+1D bulk and the usual 3D plasma variables arise as operational brane observables defined by a projection map. The central objective is not to discard MHD, but to place it inside a larger, computable hierarchy that (i) reproduces MHD in controlled limits and (ii) predicts explicit, simulation-resolvable beyond-MHD correction channels when those limits are relaxed.

What was constructed. The paper defined a coupled bulk system consisting of a localized electromagnetic sector (Section 3) and a multi-species plasma matter sector (Section 4), collected as a single simulation target (Section 5). The electromagnetic localization profile $Z(w)$ controls how gauge dynamics is concentrated near the operational brane, while an independent projection kernel $W(w)$ defines what the brane observer measures (Section 2). This separation is essential: the bulk system can remain conservative while the brane subsystem becomes an open system with explicit exchange terms.

Controlled recovery of standard limits. We established a controlled reduction chain

$$(\text{bulk 4+1 system}) \rightarrow (\text{effective 3+1 Maxwell} + \text{two-fluid}) \rightarrow (\text{MHD}),$$

under explicit assumptions that suppress transverse structure: zero-mode dominance, negligible mixed components $F_{\mu w}$, negligible transverse current J^w , and quasi-neutral two-fluid ordering (Section 6). In this limit, the localized Maxwell sector reduces to ordinary 3+1 Maxwell theory with an effective coupling $\mu_0^{\text{eff}} = \mu_0 / \int Z(w)dw$, and the standard ideal-MHD induction equation follows. The reduction is not merely conceptual: it is designed to be algebraically checkable and compatible with code verification.

Beyond MHD as explicit 4D interaction channels. The principal scientific output is a clean organization of the leading extensions that appear when the controlled assumptions are relaxed (Section 7):

- a dynamical scalar-photon sector A_w and mixed field components $F_{\mu w}$, introducing additional force channels and transverse electric structure;
- explicit brane–bulk exchange through $J^w \neq 0$, producing leakage/source terms in projected brane balance laws even when bulk conservation is exact;
- finite-localization transverse-mode corrections (Hermite/KK tower for Gaussian Z), which provide additional storage and transport channels for energy and topology and yield fixed-coefficient Yukawa corrections in static limits;
- projected unresolved-stress contributions (“topology stress” terms) that quantify how sub- w structure manifests as effective non-ideality in brane-facing equations.

Together, these effects provide a principled alternative to ad hoc resistivity for topology-changing dynamics: brane reconnection-like behavior can be interpreted as conservative transport into transverse degrees of freedom rather than local dissipation into heat.

A concrete computational roadmap. We described two complementary simulation architectures (Section 9): direct 4D-grid evolution in (x, y, z, w) for reference and ground truth, and a 3D×modes solver that represents transverse structure in a truncated Hermite basis. The mode solver is the recommended route for massive runs because the linear transverse operator

becomes diagonal in mode index (fixed masses $m_n^2 \propto n$), enabling strong mode-parallel scaling, while nonlinear couplings are handled via pseudo-spectral reconstruction/projection in w . The diagnostic program emphasizes bulk energy closure, mode-energy spectra, and explicit leakage measures $S_{\text{leak}}^{(s)}$ to distinguish physical 4D-enabled non-ideality from numerical diffusion.

Validation and near-term targets. We outlined a benchmark program (Section 10) that (i) reproduces standard Maxwell/two-fluid/MHD behavior in the controlled limit, and (ii) isolates beyond-MHD signals through targeted activation of the A_w sector, transverse currents, and transverse modes. The near-term implementation targets are therefore clear:

1. implement the 3D×modes Maxwell update with generalized Lorenz gauge control and verify the Coulomb-plus-Yukawa tower response;
2. couple to a two-fluid or GNLS plasma module with charge-conserving current construction;
3. reproduce controlled-limit MHD benchmarks with $N_w = 1$, then increase N_w and quantify how reconnection proxies correlate with energy transfer into mixed fields and higher modes;
4. demonstrate convergence in both (x, y, z) resolution and transverse truncation (N_w , quadrature order), with bulk energy ledger closure as a primary correctness gate.

Outlook. The framework presented here provides a coherent hierarchy: a conservative bulk interaction model, an operational brane-observer map, a controlled MHD limit, and a structured set of beyond-MHD correction channels with explicit simulation knobs. Immediate extensions include: (i) incorporating relativistic kinetic matter for extreme regimes, (ii) adding collision and radiation models consistent with bulk conservation, and (iii) allowing back-reaction of plasma loads on localization/geometry degrees of freedom when those sectors are activated. The key promise is that reconnection and other topology-changing plasma processes become diagnosable consequences of evolving a larger interaction system, rather than exceptions that require phenomenological fixes at the outset.

A Notation Quick Reference

This appendix lists the most frequently used symbols and conventions.

Indices and coordinates

Symbol	Meaning
x^M	Bulk spacetime coordinates: $x^M = (t, x^1, x^2, x^3, w) \equiv (t, \mathbf{x}, w)$
$\mathbf{x}$	Brane spatial coordinates: $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$
$\mathbf{X}$	Bulk spatial coordinates: $\mathbf{X} = (x, y, z, w) \in \mathbb{R}^4$
$M, N, \dots$	Bulk spacetime indices in $\{0, 1, 2, 3, 4\}$ with $0 \equiv t$, $4 \equiv w$
$\mu, \nu, \dots$	Brane spacetime indices in $\{0, 1, 2, 3\}$
$A, B, \dots$	Bulk spatial indices in $\{1, 2, 3, 4\}$
$a, b, c, \dots$	Brane spatial indices in $\{1, 2, 3\}$
η_{MN}	Flat bulk metric, $\eta_{MN} = \text{diag}(-1, +1, +1, +1, +1)$
ϵ_{abc}	Brane Levi-Civita symbol, $\epsilon_{xyz} = +1$

Differential operators and measures

Symbol	Meaning
∂_M	Partial derivative $\partial/\partial x^M$
∇_3	Brane gradient $(\partial_x, \partial_y, \partial_z)$
∇_4	Bulk spatial gradient (∇_3, ∂_w)
∇_3^2	Brane Laplacian $\partial_x^2 + \partial_y^2 + \partial_z^2$
∇_4^2	Bulk spatial Laplacian $\nabla_3^2 + \partial_w^2$
$\square_4$	Brane d'Alembertian $-\partial_t^2 + \nabla_3^2$
$\square_5$	Bulk d'Alembertian $-\partial_t^2 + \nabla_3^2 + \partial_w^2$
$\int d^3x$	$\int_{\mathbb{R}^3} dx dy dz$
$\int dw$	$\int_{-\infty}^{+\infty} dw$
$\int d^4X$	$\int_{\mathbb{R}^3} d^3x \int_{\mathbb{R}} dw$

Localization, projection, and effective couplings

Symbol	Meaning
$Z(w)$	EM localization profile (weight in the bulk Maxwell action)
$W(w)$	Brane projection / measurement kernel, normalized by $\int dw W(w) = 1$
$\overline{Q}$	Brane observable (projection): $\overline{Q}(t, \mathbf{x}) = \int dw W(w) Q(t, \mathbf{x}, w)$
$\mathcal{Z}_{\text{int}}$	Localization integral $\mathcal{Z}_{\text{int}} = \int dw Z(w)$
μ_0^{eff}	Effective brane coupling $\mu_0^{\text{eff}} = \mu_0 / \mathcal{Z}_{\text{int}}$ (controlled brane Maxwell limit)
ϵ_0^{eff}	Effective permittivity satisfying $\epsilon_0^{\text{eff}} \mu_0^{\text{eff}} = c^{-2}$
λ	Localization width (Gaussian example: $Z(w) = e^{-w^2/\lambda^2}$)

Electromagnetic sector (4+1D)

Symbol	Meaning
A_M	Bulk gauge potential $A_M = (A_0, A_a, A_w)$
F_{MN}	Field strength $F_{MN} = \partial_M A_N - \partial_N A_M$
J^M	Total bulk 5-current $J^M = (J^0, J^a, J^w)$ (sources Maxwell)
ξ	Gauge-fixing parameter (generalized Lorenz gauge)
E_A	Bulk electric components $E_A \equiv F_{A0}$, with $A \in \{x, y, z, w\}$
E_w	Transverse electric component $E_w \equiv F_{w0} = -\partial_t A_w - \partial_w A_0$
C_a	Mixed components $C_a \equiv F_{aw} = \partial_a A_w - \partial_w A_a$
$\mathbf{E}^{(\text{brane})}$	Brane electric field: $E_a^{(\text{brane})} \equiv \overline{F_{a0}}$
$\mathbf{B}^{(\text{brane})}$	Brane magnetic field: $B_a^{(\text{brane})} \equiv \frac{1}{2} \epsilon_{abc} \overline{F_{bc}}$
ρ_q	Brane charge density (in the controlled 3 + 1 Maxwell limit)
$\mathbf{J}$	Brane current density (in the controlled 3 + 1 Maxwell limit)

Plasma matter sector (bulk and brane observables)

Symbol	Meaning
s	Species index (e.g. $s \in \{e, i\}$ for electrons/ions)
m_s, q_s	Mass and charge of species s
ψ_s	Species field (GNLS option)
f_s	Species distribution function (Vlasov/PIC option)
ρ_s	Bulk species number density (GNLS: $\rho_s = \psi_s ^2$; Vlasov: $\rho_s = \int f_s d^4v$)
j_s^A	Bulk species number current ($A \in \{x, y, z, w\}$)
v_s^A	Bulk species velocity, $v_s^A \equiv j_s^A / \rho_s$ (when $\rho_s > 0$)
J_s^M	Species charge 5-current: $J_s^0 = q_s \rho_s$, $J_s^A = q_s j_s^A$
$\bar{\rho}_s$	Brane-observed species density $\bar{\rho}_s = \int dw W \rho_s$
$\bar{j}_s^a$	Brane-observed species current $\bar{j}_s^a = \int dw W j_s^a$
$\mathbf{v}_{s,\text{brane}}$	Brane-observed species velocity $\mathbf{v}_{s,\text{brane}} = \bar{\mathbf{j}}_s / \bar{\rho}_s$
$S_{\text{leak}}^{(s)}$	Leakage/source in projected continuity: $\partial_t \bar{\rho}_s + \partial_a \bar{j}_s^a = S_{\text{leak}}^{(s)}$
$U_s(\rho_s), h_s(\rho_s)$	EOS potential and enthalpy-like function: $h_s = dU_s/d\rho_s$
Q_s	Dispersive/quantum potential $Q_s = -(\hbar^2/2m_s)(\nabla_4^2 \sqrt{\rho_s})/\sqrt{\rho_s}$ (GNLS option)

Two-fluid and MHD variables (controlled brane limit)

Symbol	Meaning
n_s	Brane species number density (controlled limit, with $s \in \{e, i\}$)
$\mathbf{v}_s$	Brane species velocity (3-vector)
ρ	Single-fluid mass density $\rho = m_i n_i + m_e n_e$
$\mathbf{v}$	Center-of-mass (single-fluid) velocity on the brane (3-vector)
$\mathbf{J}$	Brane current density, $\mathbf{J} = en(\mathbf{v}_i - \mathbf{v}_e)$ (quasi-neutral)
p_s, p	Species pressures and total pressure $p = p_i + p_e$
η	(Optional) resistivity in resistive closures, $\mathbf{E} + \mathbf{v} \times \mathbf{B} = \eta \mathbf{J} + \dots$

Transverse-mode (Hermite/KK) notation

Symbol	Meaning
$f_n(w), \phi_n(w)$	Transverse basis functions (Gaussian case: Hermite family)
$a_\mu^{(n)}(t, \mathbf{x})$	Brane mode amplitude of A_μ in the expansion $A_\mu = \sum_n a_\mu^{(n)} f_n(w)$
m_n	Mode mass, Gaussian case: $m_n^2 = 2n/\lambda^2$
c_n	Mode coupling weight to brane-localized sources (defined from $f_n(0)$ and $\ f_n\ _Z$)
N_w	Number of retained transverse modes in a truncated expansion
Q_w	Quadrature order / number of transverse nodes in pseudo-spectral projection

B Derivation Details: Maxwell Reduction and μ_0^{eff}

This appendix provides the detailed derivation of the effective 3 + 1 Maxwell sector and the appearance of the effective coupling $\mu_0^{\text{eff}} = \mu_0 / \int dw Z(w)$ used throughout the paper. We give two equivalent derivations: (i) reduction at the level of the action, and (ii) reduction at the level of the field equations. Both make clear the assumptions under which the reduction is valid and identify the precise terms that spoil it when w -structure is excited.

B.1 Localized Maxwell action in 4 + 1D

We take the bulk electromagnetic action (with an optional gauge-fixing term)

$$S_{\text{EM}}[A; Z] = \int dt d^3x dw \left[-\frac{Z(w)}{4\mu_0} F_{MN} F^{MN} - \frac{1}{2\xi\mu_0} (\partial \cdot A)^2 - A_M J^M \right], \quad (153)$$

where $M, N \in \{0, 1, 2, 3, 4\}$ with $0 \equiv t$ and $4 \equiv w$, $F_{MN} = \partial_M A_N - \partial_N A_M$, and $Z(w) \geq 0$ is integrable:

$$Z_{\text{int}} \equiv \int_{-\infty}^{+\infty} dw Z(w) < \infty. \quad (154)$$

(For the Gaussian example $Z(w) = e^{-w^2/\lambda^2}$, one has $Z_{\text{int}} = \lambda\sqrt{\pi}$.)

Dimensional note. Because $\int dw Z$ carries units of length if Z is dimensionless, the bulk constant μ_0 in (153) should be understood as the 4 + 1 coupling whose units make the action consistent; after reduction, μ_0^{eff} has the usual 3 + 1 units. (Equivalently, one may label the bulk constant $\mu_{0,5}$ and define $\mu_0^{\text{eff}} \equiv \mu_{0,5}/Z_{\text{int}}$.)

B.2 Reduction at the level of the action

We now impose the controlled brane/zero-mode assumptions used in Section 6, specialized here to the EM sector:

1. Zero-mode dominance for brane components:

$$A_\mu(t, \mathbf{x}, w) \simeq a_\mu(t, \mathbf{x}), \quad \partial_w A_\mu \simeq 0, \quad \mu \in \{0, 1, 2, 3\}. \quad (155)$$

2. Suppressed mixed sector in this limit:

$$F_{\mu w} \simeq 0. \quad (156)$$

A sufficient condition is axial gauge $A_w = 0$ together with $\partial_w A_\mu \simeq 0$.

Under (155)–(156),

$$F_{\mu\nu}(t, \mathbf{x}, w) \simeq f_{\mu\nu}(t, \mathbf{x}) \equiv \partial_\mu a_\nu - \partial_\nu a_\mu, \quad F_{\mu w} \simeq 0, \quad (157)$$

and therefore

$$F_{MN} F^{MN} \simeq f_{\mu\nu} f^{\mu\nu}. \quad (158)$$

Substituting (158) into (153) yields an effective 3 + 1 action:

$$S_{\text{EM}} \simeq \int dt d^3x \left[-\frac{1}{4\mu_0} \left(\int dw Z(w) \right) f_{\mu\nu} f^{\mu\nu} - \int dw a_\mu J^\mu(t, \mathbf{x}, w) \right] + (\text{gauge-fixing}), \quad (159)$$

$$= \int dt d^3x \left[-\frac{1}{4\mu_0^{\text{eff}}} f_{\mu\nu} f^{\mu\nu} - a_\mu j_{\text{eff}}^\mu(t, \mathbf{x}) \right] + (\text{gauge-fixing}), \quad (160)$$

where we define the effective brane current and effective coupling by

$$j_{\text{eff}}^\mu(t, \mathbf{x}) \equiv \int_{-\infty}^{+\infty} dw J^\mu(t, \mathbf{x}, w), \quad \mu_0^{\text{eff}} \equiv \frac{\mu_0}{Z_{\text{int}}}. \quad (161)$$

Brane-localized sources as a special case. If the source is strictly brane-localized, $J^\mu(t, \mathbf{x}, w) = j_b^\mu(t, \mathbf{x})\delta(w)$, then $j_{\text{eff}}^\mu = j_b^\mu$, and the reduced action is exactly the standard Maxwell action (with $\mu_0 \rightarrow \mu_0^{\text{eff}}$) coupled to the brane current.

Gauge-fixing term. If $(\partial \cdot A)$ is also w -independent in the controlled limit, the gauge-fixing term reduces with the same prefactor Z_{int} and is absorbed by the replacement $\mu_0 \rightarrow \mu_0^{\text{eff}}$ exactly as above.

B.3 Reduction at the level of the field equations

Varying (153) gives the localized Maxwell equations

$$\partial_M (Z(w) F^{MN}) + \frac{1}{\xi} \partial^N (\partial \cdot A) = \mu_0 J^N. \quad (162)$$

We focus on the brane components $N = \nu \in \{0, 1, 2, 3\}$ and integrate over w :

$$\int dw \partial_M (Z F^{M\nu}) + \frac{1}{\xi} \int dw \partial^\nu (\partial \cdot A) = \mu_0 \int dw J^\nu. \quad (163)$$

Split the first term into μ and w contributions and integrate by parts in w :

$$\partial_\mu \left(\int dw Z F^{\mu\nu} \right) + [Z F^{w\nu}]_{-\infty}^{+\infty} + \frac{1}{\xi} \partial^\nu \left(\int dw (\partial \cdot A) \right) = \mu_0 j_{\text{eff}}^\nu. \quad (164)$$

Assuming localized fields such that the boundary flux term vanishes, $[Z F^{w\nu}]_{-\infty}^{+\infty} = 0$, and invoking the controlled-limit suppression $F^{w\nu} \simeq 0$ and w -independence $F^{\mu\nu} \simeq f^{\mu\nu}$, we obtain

$$\partial_\mu \left(\int dw Z \right) f^{\mu\nu} + \frac{1}{\xi} \partial^\nu (\partial_\mu a^\mu) \left(\int dw Z \right) = \mu_0 j_{\text{eff}}^\nu. \quad (165)$$

Dividing by Z_{int} yields the standard 3 + 1 Maxwell equation with effective coupling

$$\partial_\mu f^{\mu\nu} + \frac{1}{\xi} \partial^\nu (\partial_\mu a^\mu) = \mu_0^{\text{eff}} j_{\text{eff}}^\nu, \quad \mu_0^{\text{eff}} = \frac{\mu_0}{Z_{\text{int}}}. \quad (166)$$

Where the reduction fails. Equation (164) makes the failure modes explicit:

- If $F^{w\nu}$ is not negligible, then the boundary/flux term and the w -mixing in $Z F^{w\nu}$ re-enter and the brane sector is not closed.
- If $F^{\mu\nu}$ has significant w -dependence in the support of Z , then $\int dw Z F^{\mu\nu} \neq Z_{\text{int}} f^{\mu\nu}$ and higher transverse modes contribute.

These are precisely the channels that generate beyond-MHD corrections in the brane-facing plasma equations.

B.4 Alternative viewpoint: KK/Hermite normalization and the zero mode

For Gaussian localization (and more generally for any integrable Z), the transverse Sturm–Liouville problem defines an orthogonal basis $\{f_n(w)\}$ with weight Z . Writing a mode expansion for the brane components,

$$A_\mu(t, \mathbf{x}, w) = \sum_{n \geq 0} a_\mu^{(n)}(t, \mathbf{x}) f_n(w), \quad (167)$$

and inserting into the quadratic action produces a sum of decoupled kinetic terms with coefficients given by the weighted norms $\|f_n\|_Z^2 = \int dw Z f_n^2$.

The massless (Maxwell) mode is the constant mode $f_0(w) = 1$, whose norm is

$$\|f_0\|_Z^2 = \int dw Z(w) = Z_{\text{int}}. \quad (168)$$

Thus the kinetic term for $a_\mu^{(0)}$ is

$$-\frac{1}{4\mu_0}\|f_0\|_Z^2 f_{\mu\nu}f^{\mu\nu} = -\frac{1}{4\mu_0^{\text{eff}}} f_{\mu\nu}f^{\mu\nu}, \quad \mu_0^{\text{eff}} = \frac{\mu_0}{\|f_0\|_Z^2} = \frac{\mu_0}{Z_{\text{int}}}, \quad (169)$$

which reproduces the result from the action/equation reductions. From this perspective, μ_0^{eff} is simply the bulk coupling rescaled by the normalization of the zero mode.

B.5 Connection to ϵ_0^{eff} and the speed of light

If one wishes to retain a fixed brane light speed c in SI form, then the effective permittivity is defined by

$$\epsilon_0^{\text{eff}} \mu_0^{\text{eff}} = \frac{1}{c^2}. \quad (170)$$

In practice, once μ_0^{eff} is fixed by the localization integral Z_{int} , this equation defines ϵ_0^{eff} . In natural units $c = 1$, the relation is $\epsilon_0^{\text{eff}} = 1/\mu_0^{\text{eff}}$.

Operational note. The reduction in this appendix concerns the *effective* brane Maxwell sector sourced by the w -integrated current j_{eff}^μ . Separately, brane-measured fields in the main text are defined by the operational projection kernel $W(w)$ via $\bar{Q} = \int dw WQ$. These are distinct operations (reduction vs. measurement) and generally coincide only under the controlled-limit assumptions where w -dependence is negligible.

C Derivation Details: Two-Fluid $\rightarrow$ MHD

This appendix records the standard algebraic reduction from a brane two-fluid plasma (electrons + ions) to single-fluid MHD. The only modification relative to textbook SI-MHD is that, in our framework, μ_0 may be replaced by μ_0^{eff} in Ampère's law if one is working in the controlled brane-Maxwell limit of Appendix B. Otherwise the reduction is unchanged.

Throughout, all fields and fluid variables in this appendix are *brane* 3-vectors and 3-scalar densities; we use ∇_3 as the brane gradient.

C.1 Starting point: brane two-fluid + Maxwell

Consider two charged species: ions (i) and electrons (e):

$$q_i = +e, \quad q_e = -e. \quad (171)$$

Let $n_s(t, \mathbf{x})$ be the number density, $\mathbf{v}_s(t, \mathbf{x})$ the species velocity, and $p_s(t, \mathbf{x})$ the species pressure.

Continuity (each species).

$$\partial_t n_s + \nabla_3 \cdot (n_s \mathbf{v}_s) = 0, \quad s \in \{i, e\}. \quad (172)$$

Momentum (each species).

$$m_s n_s \left(\partial_t + \mathbf{v}_s \cdot \nabla_3 \right) \mathbf{v}_s = q_s n_s \left(\mathbf{E} + \mathbf{v}_s \times \mathbf{B} \right) - \nabla_3 p_s + \mathbf{R}_s, \quad s \in \{i, e\}. \quad (173)$$

Here $\mathbf{R}_s$ is the interspecies momentum-exchange (collisional) term, taken to satisfy total momentum conservation

$$\mathbf{R}_i + \mathbf{R}_e = \mathbf{0}. \quad (174)$$

Maxwell (brane). We use the standard brane Maxwell equations:

$$\nabla_3 \cdot \mathbf{B} = 0, \quad \nabla_3 \times \mathbf{E} = -\partial_t \mathbf{B}, \quad (175)$$

$$\nabla_3 \cdot \mathbf{E} = \frac{\rho_q}{\epsilon_0^{\text{eff}}}, \quad \nabla_3 \times \mathbf{B} - \frac{1}{c^2} \partial_t \mathbf{E} = \mu_0^{\text{eff}} \mathbf{J}. \quad (176)$$

If one is not in the controlled reduction regime, replace $\epsilon_0^{\text{eff}} \rightarrow \epsilon_0$ and $\mu_0^{\text{eff}} \rightarrow \mu_0$.

The brane charge density and current are

$$\rho_q \equiv e(n_i - n_e), \quad \mathbf{J} \equiv e(n_i \mathbf{v}_i - n_e \mathbf{v}_e). \quad (177)$$

C.2 Single-fluid variables and inversion identities

Define the total mass density ρ , center-of-mass velocity $\mathbf{v}$, and total pressure p by

$$\rho \equiv m_i n_i + m_e n_e, \quad \mathbf{v} \equiv \frac{m_i n_i \mathbf{v}_i + m_e n_e \mathbf{v}_e}{\rho}, \quad p \equiv p_i + p_e. \quad (178)$$

Quasi-neutral approximation. For MHD-like regimes we adopt quasi-neutrality:

$$n_i \simeq n_e \equiv n, \quad \rho_q \simeq 0. \quad (179)$$

Then

$$\rho \simeq (m_i + m_e)n, \quad \mathbf{J} = en(\mathbf{v}_i - \mathbf{v}_e). \quad (180)$$

Exact inversion for $\mathbf{v}_i$ and $\mathbf{v}_e$ (quasi-neutral). Solving (178) and (180) for the species velocities gives

$$\mathbf{v}_i = \mathbf{v} + \frac{m_e}{m_i + m_e} \frac{\mathbf{J}}{en}, \quad \mathbf{v}_e = \mathbf{v} - \frac{m_i}{m_i + m_e} \frac{\mathbf{J}}{en}. \quad (181)$$

In the electron-inertia-suppressed limit $m_e/m_i \ll 1$,

$$\mathbf{v}_i \approx \mathbf{v}, \quad \mathbf{v}_e \approx \mathbf{v} - \frac{\mathbf{J}}{en}. \quad (182)$$

C.3 Mass continuity

Multiply (172) by m_s and sum over species:

$$\begin{aligned} \partial_t \rho + \nabla_3 \cdot (m_i n_i \mathbf{v}_i + m_e n_e \mathbf{v}_e) &= 0, \\ \Rightarrow \partial_t \rho + \nabla_3 \cdot (\rho \mathbf{v}) &= 0. \end{aligned} \quad (183)$$

This is the standard single-fluid continuity equation.

C.4 Single-fluid momentum equation

Add the ion and electron momentum equations (173). Using (174), the collisional exchange cancels. The electric forces also cancel because $q_i n_i + q_e n_e = \rho_q \simeq 0$ under quasi-neutrality. The magnetic forces combine into $\mathbf{J} \times \mathbf{B}$:

$$\begin{aligned} q_i n_i (\mathbf{v}_i \times \mathbf{B}) + q_e n_e (\mathbf{v}_e \times \mathbf{B}) &= en_i (\mathbf{v}_i \times \mathbf{B}) - en_e (\mathbf{v}_e \times \mathbf{B}) \\ &= (en_i \mathbf{v}_i - en_e \mathbf{v}_e) \times \mathbf{B} = \mathbf{J} \times \mathbf{B}. \end{aligned} \quad (184)$$

Thus, the summed momentum equation becomes

$$m_i n_i (\partial_t + \mathbf{v}_i \cdot \nabla_3) \mathbf{v}_i + m_e n_e (\partial_t + \mathbf{v}_e \cdot \nabla_3) \mathbf{v}_e = \mathbf{J} \times \mathbf{B} - \nabla_3 p. \quad (185)$$

MHD closure on inertial terms. Expressing the left-hand side of (185) exactly in terms of ρ and $\mathbf{v}$ introduces additional terms involving the relative drift $\mathbf{v}_i - \mathbf{v}_e \propto \mathbf{J}$. In standard MHD ordering these “drift-stress” corrections are of higher order and are neglected, yielding the usual single-fluid momentum equation:

$$\rho(\partial_t + \mathbf{v} \cdot \nabla_3) \mathbf{v} = \mathbf{J} \times \mathbf{B} - \nabla_3 p. \quad (186)$$

(If desired, the neglected terms can be retained and written as the divergence of a current-related stress tensor; this is one pathway to extended-MHD corrections.)

C.5 Generalized Ohm’s law

Start from the electron momentum equation (173):

$$m_e n_e (\partial_t + \mathbf{v}_e \cdot \nabla_3) \mathbf{v}_e = -en_e (\mathbf{E} + \mathbf{v}_e \times \mathbf{B}) - \nabla_3 p_e + \mathbf{R}_e. \quad (187)$$

Solve for $\mathbf{E}$:

$$\mathbf{E} + \mathbf{v}_e \times \mathbf{B} = -\frac{1}{en_e} \nabla_3 p_e + \frac{1}{en_e} \mathbf{R}_e - \frac{m_e}{e} (\partial_t + \mathbf{v}_e \cdot \nabla_3) \mathbf{v}_e. \quad (188)$$

Resistive closure (optional). A standard electron-ion drag model is

$$\mathbf{R}_e = m_e n \nu_{ei} (\mathbf{v}_i - \mathbf{v}_e), \quad (189)$$

which, under quasi-neutrality, becomes

$$\frac{1}{en} \mathbf{R}_e = \frac{m_e \nu_{ei}}{e^2 n} \mathbf{J} \equiv \eta \mathbf{J}, \quad \eta \equiv \frac{m_e \nu_{ei}}{e^2 n}. \quad (190)$$

Transform from $\mathbf{v}_e$ to $\mathbf{v}$. Using $\mathbf{v}_e \approx \mathbf{v} - \mathbf{J}/(en)$ from (182),

$$\begin{aligned} \mathbf{E} + \mathbf{v} \times \mathbf{B} &= (\mathbf{E} + \mathbf{v}_e \times \mathbf{B}) + (\mathbf{v} - \mathbf{v}_e) \times \mathbf{B} \\ &= (\mathbf{E} + \mathbf{v}_e \times \mathbf{B}) + \frac{\mathbf{J}}{en} \times \mathbf{B}. \end{aligned} \quad (191)$$

Substituting (188) and (190) yields the generalized Ohm law in the standard $(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ form:

$$\boxed{\mathbf{E} + \mathbf{v} \times \mathbf{B} = \underbrace{\eta \mathbf{J}}_{\text{resistive}} + \underbrace{\frac{\mathbf{J} \times \mathbf{B}}{en}}_{\text{Hall}} - \underbrace{\frac{1}{en} \nabla_3 p_e}_{\text{electron pressure}} - \underbrace{\frac{m_e}{e} (\partial_t + \mathbf{v}_e \cdot \nabla_3) \mathbf{v}_e}_{\text{electron inertia}}}. \quad (192)$$

Dropping the electron-inertia term yields the Hall-resistive form used most commonly in extended MHD.

Ideal MHD (Ohm). The ideal-MHD limit is the controlled sub-limit in which the right-hand side of (192) is negligible:

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B}. \quad (193)$$

C.6 Induction equation

Faraday’s law (175) implies

$$\partial_t \mathbf{B} = -\nabla_3 \times \mathbf{E}. \quad (194)$$

Insert the generalized Ohm law (192) to obtain

$$\begin{aligned} \partial_t \mathbf{B} &= \nabla_3 \times (\mathbf{v} \times \mathbf{B}) - \nabla_3 \times (\eta \mathbf{J}) - \nabla_3 \times \left(\frac{\mathbf{J} \times \mathbf{B}}{en} \right) + \nabla_3 \times \left(\frac{1}{en} \nabla_3 p_e \right) \\ &\quad + \nabla_3 \times \left[\frac{m_e}{e} (\partial_t + \mathbf{v}_e \cdot \nabla_3) \mathbf{v}_e \right]. \end{aligned} \quad (195)$$

Ideal induction equation. Using (193), (194) becomes

$$\partial_t \mathbf{B} = \nabla_3 \times (\mathbf{v} \times \mathbf{B}), \quad \nabla_3 \cdot \mathbf{B} = 0, \quad (196)$$

which is the standard ideal-MHD flux-freezing statement.

C.7 Closing the MHD system

A minimal ideal-MHD closure consists of:

$$\partial_t \rho + \nabla_3 \cdot (\rho \mathbf{v}) = 0, \quad (197)$$

$$\rho \left(\partial_t + \mathbf{v} \cdot \nabla_3 \right) \mathbf{v} = \mathbf{J} \times \mathbf{B} - \nabla_3 p, \quad (198)$$

$$\partial_t \mathbf{B} = \nabla_3 \times (\mathbf{v} \times \mathbf{B}), \quad \nabla_3 \cdot \mathbf{B} = 0, \quad (199)$$

supplemented by an equation of state (and, if needed, an energy equation). A common additional closure in the MHD ordering is to neglect displacement current in Ampère's law (176):

$$\mathbf{J} \simeq \frac{1}{\mu_0^{\text{eff}}} \nabla_3 \times \mathbf{B}, \quad (200)$$

thereby eliminating $\mathbf{J}$ in favor of $\mathbf{B}$. In resistive MHD, one keeps $-\nabla_3 \times (\eta \mathbf{J})$ in (195) and uses (200) to write it as a magnetic diffusion term.

Where the present framework modifies this appendix. The algebra above is the standard two-fluid $\rightarrow$ MHD reduction on a 3D domain. In the full $4+1$ theory, additional terms enter $\mathbf{E}$ and the brane balances when J^w , $F_{\mu w}$, and transverse-mode excitations are not negligible. Those effects appear as explicit beyond-MHD corrections in the main text and do not alter the internal consistency of the classical reduction recorded here.

D Mode Decomposition in w (Hermite / Soft-Wall)

This appendix collects the transverse (w) mode machinery induced by the localized Maxwell kinetic term. The practical purpose is twofold: (i) to make the *finite-localization corrections* computable and interpretable (Coulomb + Yukawa tower), and (ii) to provide the mathematical foundation for the $3D \times \text{modes}$ HPC algorithm described in Section 9.

D.1 Sturm–Liouville problem from localized Maxwell

Consider the localized Maxwell sector in the bulk,

$$\partial_M (Z(w) F^{MN}) + \frac{1}{\xi} \partial^N (\partial \cdot A) = \mu_0 J^N, \quad F_{MN} = \partial_M A_N - \partial_N A_M, \quad M, N \in \{0, 1, 2, 3, 4\}, \quad (201)$$

with $Z(w) \geq 0$ integrable and time-independent. In a source-free region and with a hyperbolic gauge choice (e.g. generalized Lorenz gauge), the brane components A_μ ($\mu \in \{0, 1, 2, 3\}$) separate into a brane wave operator and a transverse operator acting on w :

$$\square_4 A_\mu + \frac{1}{Z(w)} \frac{d}{dw} \left(Z(w) \frac{dA_\mu}{dw} \right) = 0, \quad \square_4 \equiv -\partial_t^2 + \nabla_3^2. \quad (202)$$

This motivates a mode expansion of the form

$$A_\mu(x, w) = \sum_{n \geq 0} a_\mu^{(n)}(x) f_n(w), \quad x \equiv (t, \mathbf{x}), \quad (203)$$

where the transverse profiles $f_n(w)$ are eigenfunctions of the weighted Sturm–Liouville operator:

$$-\frac{d}{dw} \left(Z(w) \frac{df_n}{dw} \right) = m_n^2 Z(w) f_n(w). \quad (204)$$

The natural inner product and norm are the ones induced by the action:

$$\langle f, g \rangle_Z \equiv \int_{-\infty}^{+\infty} dw Z(w) f(w) g(w), \quad \|f_n\|_Z^2 \equiv \langle f_n, f_n \rangle_Z. \quad (205)$$

With appropriate boundary/localization conditions (e.g. sufficiently fast decay so that boundary terms vanish), the operator in (204) is self-adjoint with respect to $\langle \cdot, \cdot \rangle_Z$, implying orthogonality of distinct modes: $\langle f_n, f_m \rangle_Z \propto \delta_{nm}$.

D.2 Gaussian soft-wall and the Hermite spectrum

A particularly convenient “soft-wall” choice is a Gaussian localization profile,

$$Z(w) = \exp\left(-\frac{w^2}{\lambda^2}\right), \quad (206)$$

where λ is the localization width.

Writing $y \equiv w/\lambda$, equation (204) becomes

$$\frac{d^2 f_n}{dy^2} - 2y \frac{df_n}{dy} + (m_n^2 \lambda^2) f_n = 0, \quad (207)$$

which is the Hermite differential equation. A complete set of polynomial solutions is given by the physicists’ Hermite polynomials:

$$f_n(w) = H_n\left(\frac{w}{\lambda}\right), \quad m_n^2 = \frac{2n}{\lambda^2}, \quad n = 0, 1, 2, \dots \quad (208)$$

The $n = 0$ mode is massless and corresponds to the standard 3 + 1 Maxwell field on the brane. The $n \geq 1$ modes behave as a tower of massive 3 + 1 Proca-type modes, with masses set by the localization scale λ .

Normalized basis (optional). It is often convenient to work with Z -orthonormal functions $\{\phi_n\}$ rather than raw polynomials:

$$\phi_n(w) \equiv \frac{1}{\sqrt{\|f_n\|_Z^2}} f_n(w) = \frac{H_n(w/\lambda)}{\sqrt{\lambda \sqrt{\pi} 2^n n!}}, \quad \langle \phi_n, \phi_m \rangle_Z = \delta_{nm}, \quad (209)$$

using the norm computed in (210) below.

D.3 Mode normalization and brane coupling weights

For the Gaussian soft-wall (206), the Hermite orthogonality integral gives

$$\|f_n\|_Z^2 = \int_{-\infty}^{+\infty} dw \exp\left(-\frac{w^2}{\lambda^2}\right) H_n\left(\frac{w}{\lambda}\right)^2 = \lambda \sqrt{\pi} 2^n n!. \quad (210)$$

If the bulk current is brane-localized, $J^\mu(x, w) = J_b^\mu(x) \delta(w)$, then the n -th mode is sourced in proportion to its value at $w = 0$. A convenient coupling weight is

$$c_n \equiv \frac{f_n(0)^2}{\|f_n\|_Z^2}. \quad (211)$$

This weight appears directly in the effective brane-to-brane propagator and in static potentials.

Parity selection rule. For Hermite polynomials, $H_n(0) = 0$ for odd n . Therefore

$$c_{2m+1} = 0, \quad m = 0, 1, 2, \dots, \quad (212)$$

so odd transverse modes do *not* couple to strictly brane-localized sources at the symmetry point $w = 0$. Physically this follows from: (i) $Z(w)$ is even, (ii) the brane sits at the symmetry point, and (iii) odd transverse profiles integrate to zero against $\delta(w)$.

For even $n = 2m$, the closed-form value $H_{2m}(0) = (-1)^m (2m)!/m!$ yields

$$c_{2m} = \frac{H_{2m}(0)^2}{\lambda\sqrt{\pi} 2^{2m} (2m)!} = \frac{1}{\lambda\sqrt{\pi}} \frac{1}{4^m} \binom{2m}{m}. \quad (213)$$

In particular,

$$c_0 = \frac{1}{\lambda\sqrt{\pi}} = \frac{1}{Z_{\text{int}}}, \quad Z_{\text{int}} \equiv \int_{-\infty}^{+\infty} dw Z(w) = \lambda\sqrt{\pi}, \quad (214)$$

which is the same normalization that produces $\mu_0^{\text{eff}} = \mu_0/Z_{\text{int}}$ in the controlled Maxwell reduction (Appendix B).

D.4 Effective 3 + 1 field equations in mode form

Insert the expansion (203) into the brane component equation with sources and project onto mode n by multiplying by $Z(w)f_n(w)$ and integrating over w . Using orthogonality and assuming boundary terms vanish, one obtains

$$(\square_4 + m_n^2) a_\mu^{(n)}(x) = \mu_0 J_\mu^{(n)}(x) + (\text{gauge-driver terms}), \quad (215)$$

where the projected current is

$$J_\mu^{(n)}(x) \equiv \frac{1}{\|f_n\|_Z^2} \int_{-\infty}^{+\infty} dw Z(w) f_n(w) J_\mu(x, w). \quad (216)$$

For a brane-localized current $J_\mu(x, w) = J_{b,\mu}(x)\delta(w)$,

$$J_\mu^{(n)}(x) = \frac{f_n(0)}{\|f_n\|_Z^2} J_{b,\mu}(x), \quad (217)$$

so the brane-to-brane response involves $f_n(0)^2/\|f_n\|_Z^2 = c_n$.

Interpretation. Equation (215) shows that the localized 4 + 1 Maxwell system induces a *tower* of 3 + 1 fields: the $n = 0$ field is ordinary Maxwell; each $n \geq 1$ mode behaves like a massive vector field with mass m_n . In simulations, one truncates to $n = 0, \dots, N_w - 1$ and interprets N_w as the amount of transverse structure retained.

D.5 Coulomb law and Yukawa corrections

The mode decomposition implies an effective brane-to-brane propagator

$$D_{\text{eff}}(k^2) = \sum_{n \geq 0} \frac{c_n}{k^2 - m_n^2 + i\epsilon}, \quad m_n^2 = \frac{2n}{\lambda^2}, \quad (218)$$

where k^2 is the brane Lorentz scalar built from $(\omega, \mathbf{k})$.

In the static limit ($\omega = 0$), each massive mode contributes a Yukawa potential. For a point charge q on the brane ($J_b^0 = q\delta^{(3)}(\mathbf{x})$), the scalar potential on the brane takes the form

$$A_0(r) = \frac{\mu_0 q}{4\pi r} \sum_{n \geq 0} c_n e^{-m_n r}, \quad r \equiv |\mathbf{x}|. \quad (219)$$

Separating the zero mode and using $\mu_0^{\text{eff}} = \mu_0 c_0$, one can write

$$A_0(r) = \frac{\mu_0^{\text{eff}} q}{4\pi r} \left[1 + \sum_{m \geq 1} \frac{c_{2m}}{c_0} e^{-m_{2m} r} \right], \quad m_{2m} = \frac{2\sqrt{m}}{\lambda}, \quad (220)$$

with coefficient ratios

$$\frac{c_{2m}}{c_0} = \frac{1}{4^m} \binom{2m}{m} = 1, \frac{1}{2}, \frac{3}{8}, \frac{5}{16}, \dots \quad (m = 0, 1, 2, 3, \dots). \quad (221)$$

Therefore the leading deviation from Coulomb at large r is

$$A_0(r) = \frac{\mu_0^{\text{eff}} q}{4\pi r} \left[1 + \frac{1}{2} e^{-2r/\lambda} + \mathcal{O}(e^{-2\sqrt{2}r/\lambda}) \right]. \quad (222)$$

Brane interpretation. For $r \gg \lambda$, brane observers see ordinary Coulomb behavior with coupling μ_0^{eff} . For $r \sim \lambda$, Yukawa corrections become significant and encode finite-localization physics. In plasma applications, this identifies λ as a new physical scale controlling when transverse degrees of freedom can influence brane-facing electromagnetic interactions.

D.6 How the mode tower is used in large simulations

The $3D \times$ modes algorithm (Section 9) follows directly from (215):

1. **Evolve mode amplitudes.** For each retained $n \in \{0, \dots, N_w - 1\}$, evolve $a_M^{(n)}(t, \mathbf{x})$ using a 3D hyperbolic update (wave stencil) with a diagonal “mass” term $m_n^2 a_M^{(n)}$. The n -dependence is local, enabling mode-parallel scaling.
2. **Reconstruct fields at transverse quadrature nodes (pseudo-spectral).** Choose quadrature nodes $\{w_q\}$ adapted to the Gaussian weight and reconstruct $A_M(t, \mathbf{x}, w_q) = \sum_{n=0}^{N_w-1} a_M^{(n)}(t, \mathbf{x}) \phi_n(w_q)$.
3. **Compute nonlinear plasma sources in physical w -space.** Evaluate currents $J^M(t, \mathbf{x}, w_q)$ from the matter model (two-fluid, GNLS, or PIC).
4. **Project sources back to modes.** Compute $J^{M,(n)}(t, \mathbf{x})$ by quadrature approximations to (216). This yields the mode-forcing terms for the next Maxwell update.

What is gained by the soft-wall/Hermite choice. Unlike a hard-wall truncation (Fourier box), the Gaussian soft-wall: (i) handles $w \in \mathbb{R}$ without artificial reflecting boundaries, (ii) yields a simple spectrum $m_n^2 \propto n$, (iii) provides an exact parity selection rule (212) for brane sources, and (iv) makes linear evolution mode-diagonal, which is critical for massive simulations.

Extension to the scalar-photon sector A_w . Nothing in the Sturm–Liouville construction restricts it to A_μ alone: the same basis may be used to represent A_w and mixed components $F_{\mu w}$. In plasma runs, A_w is retained as a dynamical degree of freedom unless a controlled reduction explicitly suppresses it (Section 6); its excitation is diagnosed by mixed-sector energy and by the work channel $J^w E_w$.

E Projected Energy and Helicity Budgets with Leakage

This appendix derives the brane-facing (projected) energy and magnetic-helicity budgets, making explicit how the projection $\overline{(\cdot)}$ turns the brane subsystem into an *open* system even when the bulk evolution is conservative. The key message is:

Bulk conservation laws close on the bulk fields. Brane-facing conservation laws close only after adding explicit leakage terms (transport in w) and “projection-stress” covariances (unresolved transverse structure).

E.1 Projection identities and Reynolds decomposition in w

Recall the operational brane observable map

$$\overline{Q}(t, \mathbf{x}) \equiv \int_{-\infty}^{+\infty} dw W(w) Q(t, \mathbf{x}, w), \quad \int dw W(w) = 1, \quad (223)$$

with fixed (time-independent) kernel $W(w)$. For any sufficiently smooth Q ,

$$\overline{\partial_t Q} = \partial_t \overline{Q}, \quad \overline{\partial_a Q} = \partial_a \overline{Q}, \quad (224)$$

while for w -derivatives one has the integration-by-parts identity

$$\overline{\partial_w Q} = \left[W(w) Q \right]_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} dw W'(w) Q \equiv \left[W Q \right]_{-\infty}^{+\infty} - \overline{Q}^{(\partial W)}. \quad (225)$$

If $W(w)$ decays sufficiently fast (or is compactly supported with $W = 0$ at its endpoints), the boundary term vanishes and $\overline{\partial_w Q} = -\overline{Q}^{(\partial W)}$.

Resolved + subscale decomposition. For any quantity $Q(t, \mathbf{x}, w)$ define the fluctuation

$$Q'(t, \mathbf{x}, w) \equiv Q - \overline{Q}, \quad \overline{Q'} = 0. \quad (226)$$

Then quadratic products satisfy

$$\overline{QR} = \overline{Q} \overline{R} + \overline{Q'R'}. \quad (227)$$

The covariance $\overline{Q'R'}$ is the precise mathematical object that encodes “unresolved transverse physics” (higher modes in the Hermite/KK expansion, or fine structure in a direct w -grid).

E.2 Bulk electromagnetic energy balance (localized Maxwell)

For the localized Maxwell system (no explicit time dependence in $Z(w)$), define bulk electric and magnetic-type components

$$E_A \equiv F_{A0}, \quad F_{AB} = \partial_A A_B - \partial_B A_A, \quad A, B \in \{x, y, z, w\}. \quad (228)$$

A convenient gauge-invariant electromagnetic energy density is

$$u_{\text{EM}} \equiv \frac{Z(w)}{2\mu_0} \left(E_A E_A + \frac{1}{2} F_{AB} F_{AB} \right) = \frac{Z(w)}{2\mu_0} (|\mathbf{E}|^2 + E_w^2 + |\mathbf{B}|^2 + |\mathbf{C}|^2), \quad (229)$$

where $\mathbf{E}$ and $\mathbf{B}$ denote the brane 3-vectors $E_a = F_{a0}$, $B_a = \frac{1}{2} \epsilon_{abc} F_{bc}$, and the mixed field $\mathbf{C}$ is $C_a \equiv F_{aw}$.

A natural 4-spatial-dimensional Poynting flux is

$$S_{\text{EM}}^A \equiv \frac{Z(w)}{\mu_0} E_B F_{BA}. \quad (230)$$

In components this splits into the brane flux and the transverse (into- w) flux:

$$S_{\text{EM}}^a = \frac{Z}{\mu_0} (E_b F_{ba} - E_w C_a) = \frac{Z}{\mu_0} [(\mathbf{E} \times \mathbf{B})_a - E_w C_a], \quad (231)$$

$$S_{\text{EM}}^w = \frac{Z}{\mu_0} E_a F_{aw} = \frac{Z}{\mu_0} \mathbf{E} \cdot \mathbf{C}. \quad (232)$$

Thus, nonzero mixed structure $\mathbf{C}$ generically enables energy transport along the extra dimension.

For conservative sources (and ignoring gauge-fixing terms which do not affect gauge-invariant budgets), the localized Maxwell equations imply the bulk energy identity

$$\partial_t u_{\text{EM}} + \partial_A S_{\text{EM}}^A = -J^A E_A = -(J^a E_a + J^w E_w). \quad (233)$$

The channel $J^w E_w$ is the direct “scalar-photon work” term: when A_w is dynamical, the bulk plasma can exchange energy with the transverse electric field E_w .

E.3 Projected electromagnetic energy budget and explicit w -leakage

Multiply (233) by $W(w)$ and integrate over w . Using (224) and separating brane and w divergences yields

$$\partial_t \overline{u_{\text{EM}}} + \partial_a \overline{S_{\text{EM}}^a} = -\overline{J^A E_A} + \mathcal{L}_{\text{EM}}^{(w)}, \quad (234)$$

where the *transverse leakage power density* is

$$\begin{aligned} \mathcal{L}_{\text{EM}}^{(w)} &\equiv - \int dw W(w) \partial_w S_{\text{EM}}^w \\ &= - \left[W S_{\text{EM}}^w \right]_{-\infty}^{+\infty} + \int dw W'(w) S_{\text{EM}}^w. \end{aligned} \quad (235)$$

When the boundary term vanishes, $\mathcal{L}_{\text{EM}}^{(w)} = \int dw W' S_{\text{EM}}^w$. This is the precise analog of the species-leakage term in the projected continuity equation: even if the bulk system is conservative, the brane-measured EM energy can increase or decrease because Poynting flux crosses the measurement profile in w .

Integral form over a brane region. For a brane region $\Omega \subset \mathbb{R}^3$,

$$\frac{d}{dt} \int_{\Omega} d^3x \overline{u_{\text{EM}}} = - \int_{\partial\Omega} dS n_a \overline{S_{\text{EM}}^a} - \int_{\Omega} d^3x \overline{J^A E_A} + \int_{\Omega} d^3x \mathcal{L}_{\text{EM}}^{(w)}. \quad (236)$$

The last term is the explicit w -exchange contribution to the brane EM energy ledger.

E.4 Resolved versus projected EM energy

The quantity $\overline{u_{\text{EM}}} = \overline{u_{\text{EM}}(E_A, F_{AB})}$ is the projection of the *bulk* quadratic energy density. A brane observer, however, often constructs an energy density from *projected fields*, e.g.

$$u_{\text{EM}}^{(\text{res})}(\mathbf{x}) \equiv \frac{1}{2\mu_0^{\text{eff}}} (|\overline{\mathbf{E}}|^2 + |\overline{\mathbf{B}}|^2), \quad (237)$$

in the controlled limit where μ_0^{eff} is meaningful. These two notions differ once w -structure is present.

Using the Reynolds decomposition (226), the projected quadratic energy splits into a resolved part plus a covariance (subscale) part:

$$|\overline{\mathbf{E}}|^2 = |\overline{\mathbf{E}}|^2 + |\overline{\mathbf{E}'}|^2, \quad |\overline{\mathbf{B}}|^2 = |\overline{\mathbf{B}}|^2 + |\overline{\mathbf{B}'}|^2, \quad \text{etc.} \quad (238)$$

Thus, even in a regime where one focuses only on brane fields, the correct energy ledger includes an explicitly measurable “transverse-mode energy” reservoir given by the covariance terms. In the Hermite/KK formulation these covariance terms correspond to the energy stored in the $n \geq 1$ modes and in the mixed sector $(E_w, \mathbf{C})$.

Interpretation for reconnection diagnostics. In resistive MHD, apparent magnetic-energy loss on the brane is often attributed directly to ηJ^2 heating. Here, brane magnetic-energy loss can instead correspond to: (i) transport into w (the $\mathcal{L}_{\text{EM}}^{(w)}$ term), and/or (ii) transfer into unresolved transverse structure (covariance energy), without requiring collisional dissipation. These mechanisms are conservative in the bulk.

E.5 Magnetic helicity identity at fixed w

Fix w and define the brane vector potential $\mathbf{A}(t, \mathbf{x}, w)$ and scalar potential $\Phi(t, \mathbf{x}, w) \equiv A_0(t, \mathbf{x}, w)$. Define the brane magnetic field $\mathbf{B} \equiv \nabla_3 \times \mathbf{A}$ and brane electric field $\mathbf{E} \equiv -\nabla_3 \Phi - \partial_t \mathbf{A}$. Then the standard local helicity identity holds pointwise in w :

$$\partial_t(\mathbf{A} \cdot \mathbf{B}) + \nabla_3 \cdot (\Phi \mathbf{B} + \mathbf{E} \times \mathbf{A}) = -2 \mathbf{E} \cdot \mathbf{B}. \quad (239)$$

This identity is purely $3 + 1$ (with w acting as a parameter), and is independent of the details of the $4 + 1$ dynamics.

Gauge note. Magnetic helicity is gauge invariant on a periodic domain, or for fields satisfying standard boundary conditions (e.g. $\mathbf{B} \cdot \mathbf{n} = 0$ and fixed tangential $\mathbf{A}$ on $\partial\Omega$), or when formulated as *relative helicity* with respect to a reference field. In what follows, the boundary flux term is kept explicit.

E.6 Projected helicity budget and “helicity leakage” into transverse structure

Project (239) with $W(w)$ and integrate over w . Define the *projected helicity density* and projected helicity flux:

$$\bar{h}(\mathbf{x}) \equiv \overline{\mathbf{A} \cdot \mathbf{B}} = \int dw W(\mathbf{A} \cdot \mathbf{B}), \quad \bar{\mathbf{F}}_h(\mathbf{x}) \equiv \overline{\Phi \mathbf{B} + \mathbf{E} \times \mathbf{A}}. \quad (240)$$

Then

$$\partial_t \bar{h} + \nabla_3 \cdot \bar{\mathbf{F}}_h = -2 \overline{\mathbf{E} \cdot \mathbf{B}}. \quad (241)$$

Integrating over a brane region Ω gives

$$\frac{d}{dt} \int_{\Omega} d^3x \bar{h} = -2 \int_{\Omega} d^3x \overline{\mathbf{E} \cdot \mathbf{B}} - \int_{\partial\Omega} dS \mathbf{n} \cdot \bar{\mathbf{F}}_h. \quad (242)$$

Resolved helicity and subscale (transverse) helicity. A brane-only description typically uses projected fields $\bar{\mathbf{A}}, \bar{\mathbf{B}}, \bar{\Phi}, \bar{\mathbf{E}}$ and defines the *resolved* helicity density

$$h_{\text{res}} \equiv \bar{\mathbf{A}} \cdot \bar{\mathbf{B}}, \quad \mathbf{F}_{h,\text{res}} \equiv \bar{\Phi} \bar{\mathbf{B}} + \bar{\mathbf{E}} \times \bar{\mathbf{A}}. \quad (243)$$

Because $\bar{\mathbf{B}} = \nabla_3 \times \bar{\mathbf{A}}$ and $\bar{\mathbf{E}} = -\nabla_3 \bar{\Phi} - \partial_t \bar{\mathbf{A}}$, the resolved fields satisfy the same identity:

$$\partial_t h_{\text{res}} + \nabla_3 \cdot \mathbf{F}_{h,\text{res}} = -2 \bar{\mathbf{E}} \cdot \bar{\mathbf{B}}. \quad (244)$$

However, the projected helicity $\bar{h}$ is not equal to h_{res} when w -structure is present. Using (227), define the *subscale* (transverse-structure) helicity density

$$h_{\text{sub}} \equiv \bar{h} - h_{\text{res}} = \overline{\mathbf{A}' \cdot \mathbf{B}'}, \quad (245)$$

and the corresponding subscale helicity flux

$$\mathbf{F}_{h,\text{sub}} \equiv \bar{\mathbf{F}}_h - \mathbf{F}_{h,\text{res}} = \overline{\Phi \mathbf{B} + \mathbf{E} \times \mathbf{A}} - \bar{\Phi} \bar{\mathbf{B}} - \bar{\mathbf{E}} \times \bar{\mathbf{A}}. \quad (246)$$

Subtracting (244) from (241) yields the exact subscale helicity transfer equation:

$$\partial_t h_{\text{sub}} + \nabla_3 \cdot \mathbf{F}_{h,\text{sub}} = -2(\overline{\mathbf{E} \cdot \mathbf{B}} - \bar{\mathbf{E}} \cdot \bar{\mathbf{B}}) = -2 \overline{\mathbf{E}' \cdot \mathbf{B}'}. \quad (247)$$

Interpretation as “leakage” into transverse modes. Equation (247) isolates a mathematically precise mechanism by which brane-facing (resolved) helicity can change without invoking collisional dissipation: helicity can be transferred into transverse structure (nonzero h_{sub}) through the covariance source $\overline{\mathbf{E}'} \cdot \overline{\mathbf{B}'}$. In the Hermite/KK formulation, this covariance is exactly the contribution of excited higher transverse modes (and mixed-sector structure) to helicity budgets.

Connection to reconnection diagnostics. In ideal MHD one often expects $\mathbf{E} \cdot \mathbf{B} = 0$ (or equivalently conservation of magnetic helicity up to boundary fluxes). In the present framework, even if the *resolved* fields satisfy $\overline{\mathbf{E}} \cdot \overline{\mathbf{B}} \approx 0$, the projected quadratic $\overline{\mathbf{E}} \cdot \overline{\mathbf{B}}$ can be nonzero due to transverse structure, leading to genuine evolution of $\bar{h}$ as seen by a finite-resolution brane measurement protocol. This provides a conservative, fully diagnostic pathway for topology change: the helicity ledger closes when one includes the subscale reservoir h_{sub} and the explicit w -exchange channels in the energy ledger (Section 5 and Appendix B).

E.7 Summary: what must be recorded in simulations

For verification and for interpreting beyond-MHD behavior, simulations should record:

- the projected EM energy budget (234) including the explicit w -leakage term $\mathcal{L}_{\text{EM}}^{(w)}$ in (235);
- the partition of projected quadratic energies into resolved + covariance parts as in (238) (mode energies in the $3\text{D} \times \text{modes}$ scheme);
- the projected helicity budget (241) and its resolved/subscale decomposition (247), which directly measures transfer of topological content into transverse structure.

These ledgers are the bookkeeping backbone needed to distinguish true dissipation from conservative export into extra-dimensional degrees of freedom.

F Derivation Details: Projected Generalized Ohm’s Law and the Topology EMF

This appendix provides a short derivation of the projected generalized Ohm law used in Section 8. The logic is:

1. write the bulk (pointwise in w) electron momentum equation in $4 + 1\text{D}$,
2. solve it for the brane electric field $\mathbf{E}(t, \mathbf{x}, w)$,
3. apply the operational projection $\overline{(\cdot)}$,
4. add and subtract $\bar{\mathbf{v}} \times \bar{\mathbf{B}}$ to expose standard two-fluid/MHD terms plus the new 4D/projection corrections.

F.1 Bulk two-fluid momentum equation (electron) in $4 + 1\text{D}$

Let $A, B \in \{x, y, z, w\}$ denote bulk spatial indices and $a, b, c \in \{x, y, z\}$ denote brane spatial indices. For species s , the bulk number density and bulk velocity are $n_s(t, \mathbf{x}, w)$ and $v_s^A(t, \mathbf{x}, w)$. The bulk electromagnetic tensor is F_{MN} , with

$$E_A \equiv F_{A0}, \quad B_a \equiv \frac{1}{2} \epsilon_{abc} F_{bc}, \quad C_a \equiv F_{aw} = \partial_a A_w - \partial_w A_a. \quad (248)$$

Write the full 4D convective derivative (for electrons) as

$$D_t^{(4)} \equiv \partial_t + v_e^a \partial_a + v_e^w \partial_w. \quad (249)$$

A generic (nonrelativistic) bulk electron momentum balance has the form

$$m_e n_e D_t^{(4)} v_{e,A} = q_e n_e \left(E_A + F_{AB} v_e^B \right) - \partial_A p_e + R_{e,A} + \mathcal{F}_{e,A}^{(\text{extra})}, \quad q_e = -e, \quad (250)$$

where p_e is the electron pressure (or the equivalent enthalpy-gradient term in the GNLS realization), $R_{e,A}$ is electron-ion momentum exchange (optional), and $\mathcal{F}_{e,A}^{(\text{extra})}$ collects any additional modeled forces (e.g. dispersive terms if retained).

F.2 Mixed-sector decomposition of the Lorentz force

Restrict (250) to brane components $A = a$. Using antisymmetry of F_{AB} ,

$$F_{ab} v_e^b = (\mathbf{v}_e \times \mathbf{B})_a, \quad F_{aw} v_e^w = v_e^w C_a, \quad (251)$$

so the brane-component Lorentz term becomes

$$E_a + F_{ab} v_e^b + F_{aw} v_e^w = E_a + (\mathbf{v}_e \times \mathbf{B})_a + v_e^w C_a. \quad (252)$$

This identifies the key beyond-3+1 channel: $v_e^w C_a$, which vanishes if either $v_e^w = 0$ (no transverse drift) or $C_a = F_{aw} = 0$ (no mixed aw structure).

F.3 Pointwise (in w) bulk generalized Ohm law

Solve (250) for the brane electric field $\mathbf{E} = (E_x, E_y, E_z)$ using (252). In vector notation (all quantities still depend on w):

$$\mathbf{E} = -\mathbf{v}_e \times \mathbf{B} - v_e^w \mathbf{C} - \frac{1}{en_e} \nabla_3 p_e + \frac{1}{en_e} \mathbf{R}_e - \frac{m_e}{e} D_t^{(4)} \mathbf{v}_e + \frac{1}{en_e} \mathcal{F}_e^{(\text{extra})}, \quad (253)$$

where ∇_3 differentiates only brane coordinates $\mathbf{x} = (x, y, z)$ and $\mathbf{C} = (C_x, C_y, C_z)$.

F.4 Operational projection and exact brane-facing identity

Recall the operational brane projection

$$\overline{Q}(t, \mathbf{x}) \equiv \int_{-\infty}^{+\infty} dw W(w) Q(t, \mathbf{x}, w), \quad \int dw W(w) = 1. \quad (254)$$

Project (253) to obtain the *exact* identity

$$\overline{\mathbf{E}} = -\overline{\mathbf{v}_e \times \mathbf{B}} - \overline{v_e^w \mathbf{C}} - \overline{\frac{1}{en_e} \nabla_3 p_e} + \overline{\frac{1}{en_e} \mathbf{R}_e} - \frac{m_e}{e} \overline{D_t^{(4)} \mathbf{v}_e} + \overline{\frac{1}{en_e} \mathcal{F}_e^{(\text{extra})}}. \quad (255)$$

At this stage, no MHD closure has been assumed; the only operation performed is the measurement/projection map.

F.5 Add-subtract $\overline{\mathbf{v}} \times \overline{\mathbf{B}}$ and expose standard terms

Introduce a brane single-fluid velocity $\overline{\mathbf{v}}$ and brane current $\overline{\mathbf{J}}$ in the usual quasi-neutral ordering (Appendix C). Add and subtract $\overline{\mathbf{v}} \times \overline{\mathbf{B}}$ to (255):

$$\overline{\mathbf{E}} + \underbrace{\overline{\mathbf{v}} \times \overline{\mathbf{B}}}_{\text{Hall rewrite}} = \underbrace{\left(\overline{\mathbf{v}_e \times \mathbf{B}} - \overline{\mathbf{v}_e} \times \overline{\mathbf{B}} \right)}_{\text{projection covariance}} + \left[\text{all remaining projected terms from (255)} \right]. \quad (256)$$

Hall term (standard). Under quasi-neutrality with $\bar{n} \equiv \bar{n}_e \simeq \bar{n}_i$ and the standard ordering $m_e \ll m_i$, one uses

$$\bar{\mathbf{v}} - \bar{\mathbf{v}}_e \approx \frac{\bar{\mathbf{J}}}{e \bar{n}}, \quad (257)$$

which gives the usual Hall contribution

$$(\bar{\mathbf{v}} - \bar{\mathbf{v}}_e) \times \bar{\mathbf{B}} \approx \frac{\bar{\mathbf{J}} \times \bar{\mathbf{B}}}{e \bar{n}}. \quad (258)$$

Covariance EMF (new, exact). Define the cross-covariance EMF

$$\mathcal{E}_{\text{cov}} \equiv \bar{\mathbf{v}} \times \bar{\mathbf{B}} - \overline{\mathbf{v}_e \times \mathbf{B}} = -(\overline{\mathbf{v}_e \times \mathbf{B}} - \bar{\mathbf{v}}_e \times \bar{\mathbf{B}}). \quad (259)$$

This term vanishes if the fields have negligible w -dependence (controlled reduction), but is otherwise generically nonzero even in laminar flows.

F.6 Mixed-sector term and projection residuals

The remaining terms in (255) are grouped as follows.

Mixed-sector transverse EMF (new). The 4 + 1D mixed coupling produces the brane-facing contribution

$$\mathcal{E}_w \equiv -\overline{v_e^w \mathbf{C}}, \quad \mathbf{C} = (F_{xw}, F_{yw}, F_{zw}). \quad (260)$$

This is a direct, computable pathway for brane non-ideality sourced by transverse drift and mixed fields.

Projection residuals (exact bookkeeping). Projection does not commute with ratios or nonlinear closures. For compactness, define

$$\mathcal{E}_p \equiv -\overline{\frac{1}{en_e} \nabla_3 p_e} + \frac{1}{e \bar{n}} \nabla_3 \bar{p}_e, \quad (261)$$

$$\mathcal{E}_R \equiv \overline{\frac{1}{en_e} \mathbf{R}_e} - \eta \bar{\mathbf{J}}, \quad (262)$$

$$\mathcal{E}_I \equiv -\frac{m_e}{e} \left(\overline{D_t^{(4)} \mathbf{v}_e} - \mathcal{I}_e \right), \quad (263)$$

where $\eta \bar{\mathbf{J}}$ and $\mathcal{I}_e$ are the *chosen* brane closures for resistivity and electron inertia (if retained). In ideal-MHD-like limits these are often dropped; in Hall/extended MHD they are kept in standard forms (Appendix C).

F.7 Final projected generalized Ohm law

Collecting (258)–(263) yields the brane-facing generalized Ohm law:

$$\boxed{\bar{\mathbf{E}} + \bar{\mathbf{v}} \times \bar{\mathbf{B}} = \frac{\bar{\mathbf{J}} \times \bar{\mathbf{B}}}{e \bar{n}} - \frac{1}{e \bar{n}} \nabla_3 \bar{p}_e + \eta \bar{\mathbf{J}} - \frac{m_e}{e} \mathcal{I}_e + \mathcal{E}_{\text{topo}} + \frac{1}{en_e} \mathcal{F}_e^{(\text{extra})}}, \quad (264)$$

where the topology EMF is the sum of explicitly 4D/projection-driven contributions:

$$\boxed{\mathcal{E}_{\text{topo}} \equiv \mathcal{E}_{\text{cov}} + \mathcal{E}_w + \mathcal{E}_p + \mathcal{E}_R + \mathcal{E}_I.} \quad (265)$$

Equations (264)–(265) are equivalent to the statement in Section 8, but with the intermediate bookkeeping made explicit.

Use in the induction equation. Projected Faraday gives $\partial_t \bar{\mathbf{B}} = -\nabla_3 \times \bar{\mathbf{E}}$, so any component of $\mathcal{E}_{\text{topo}}$ with nonzero curl acts as a brane-facing non-ideal source. In particular, $\mathcal{E}_w = -\overline{v_e^w} \bar{\mathbf{C}}$ provides a conservative, mixed-sector pathway for reconnection-like behavior without inserting ad hoc resistivity, while the covariance term $\mathcal{E}_{\text{cov}}$ quantifies the direct impact of transverse structure (mode tower / unresolved w -variation) on brane topology.

G Geometry DOFs and Coupling to Plasma Loads

This appendix describes how to endow the model with a small set of *collective* geometry degrees of freedom (DOFs) and how those DOFs are driven by *plasma loads* (currents, fields, and pressure/enthalpy). The intent is computational: rather than solving a full geometric PDE, we treat the transverse localization and throat geometry as a low-dimensional subsystem coupled (in a ledger-consistent way) to the bulk plasma–EM fields. This provides a principled mechanism for (i) time-dependent localization width, (ii) dynamic tube/throat breathing, and (iii) conservative export of energy/topology into transverse/geometry channels.

G.1 Collective geometry variables

We introduce a vector of geometry variables $\Theta^I(t)$. The minimal set used in the $4 + 1$ model papers is

$$\Theta^I(t) \in \{a(t), L(t)\}, \quad (266)$$

where $a(t)$ is an effective transverse radius (in the (x, y, z) directions) and $L(t)$ is an effective extent in the transverse direction w .

For plasma applications, a second (often more directly useful) parameterization is to treat the electromagnetic localization width $\lambda(t)$ as a geometry DOF through

$$Z(w; \lambda) = \exp\left(-\frac{w^2}{\lambda^2}\right), \quad \Theta^I(t) \in \{\lambda(t)\}, \quad (267)$$

or, more generally, to allow Z to depend on Θ^I :

$$Z(w) \longrightarrow Z(w; \Theta(t)). \quad (268)$$

In the $3\text{D} \times \text{modes}$ formulation (Appendix D), λ controls the KK/Hermite spectrum $m_n^2 \propto 1/\lambda^2$, so a time-dependent $\lambda(t)$ is a direct, low-dimensional knob for transverse-mode accessibility.

Operational note. We treat the projection kernel $W(w)$ as part of the *measurement definition* and keep it fixed, even when $Z(w; \Theta)$ changes. (One can also parametrize W , but that changes what the brane observer means, rather than changing the underlying interaction geometry.)

G.2 Where geometry enters the coupled system

At the level of the action/energy ledger, geometry variables enter only through explicit parameter dependence of (i) the EM localization profile $Z(w; \Theta)$, (ii) any confinement or core-regularization potential $V_{\text{conf}}(\mathbf{X}; \Theta)$ in the matter sector, and (iii) a standalone geometry energy $E_{\text{geom}}(\Theta)$ representing wall tension, volume work, etc. Schematically:

$$S[A, \{\text{matter}\}, \Theta] = S_{\text{EM}}[A; Z(\cdot; \Theta)] + S_{\text{mat}}[\{\text{matter}\}; V_{\text{conf}}(\cdot; \Theta)] + S_{\text{geom}}[\Theta]. \quad (269)$$

A referee-clean closure for the geometry subsystem is an effective mechanical Lagrangian

$$\boxed{\mathcal{L}_{\text{geom}} = \frac{1}{2} M_{IJ} \dot{\Theta}^I \dot{\Theta}^J - E_{\text{geom}}(\Theta) - E_{\text{ratio}}(\Theta)}, \quad (270)$$

with positive-definite inertia matrix M_{IJ} . The optional penalty E_{ratio} enforces a preferred manifold (e.g. $L \simeq \alpha a$) when desired:

$$E_{\text{ratio}}(a, L) = \frac{\kappa}{2} (L - \alpha a)^2, \quad (\kappa \geq 0). \quad (271)$$

G.3 Generalized forces from the total Hamiltonian

Define the conservative total energy

$$H_{\text{tot}} = H_{\text{EM}}[A; \Theta] + H_{\text{mat}}[\{\text{matter}\}; \Theta] + E_{\text{geom}}(\Theta) + E_{\text{ratio}}(\Theta), \quad (272)$$

where H_{EM} is the bulk electromagnetic Hamiltonian (including $Z(w; \Theta)$), and H_{mat} is the bulk plasma Hamiltonian appropriate to the chosen matter model (two-fluid, GNLS, Vlasov/PIC, etc.).

The *ledger definition* of the generalized force on Θ^I is

$$F_I \equiv -\frac{\partial H_{\text{tot}}}{\partial \Theta^I}, \quad (273)$$

where the derivative is taken at fixed instantaneous fields (Hellmann–Feynman / virtual work).

With (272), the force splits into contributions:

$$F_I = F_I^{(\text{EM})} + F_I^{(\text{mat})} + F_I^{(\text{geom})} + F_I^{(\text{ratio})}. \quad (274)$$

(i) Geometry force from localized Maxwell. Using the localized Maxwell energy density (Appendix E),

$$H_{\text{EM}} = \int d^3x dw \frac{Z(w; \Theta)}{2\mu_0} \left(E_A E_A + \frac{1}{2} F_{AB} F_{AB} \right), \quad A, B \in \{x, y, z, w\}, \quad (275)$$

the explicit parametric force contribution is

$$F_I^{(\text{EM})} = -\frac{\partial H_{\text{EM}}}{\partial \Theta^I} = -\int d^3x dw \frac{\partial_{\Theta^I} Z(w; \Theta)}{2\mu_0} \left(E_A E_A + \frac{1}{2} F_{AB} F_{AB} \right) + F_{I, \text{extra}}^{(\text{EM})}. \quad (276)$$

The term $F_{I, \text{extra}}^{(\text{EM})}$ is present only if additional EM coefficients (or external-source profiles) are made geometry-dependent. If Z is the only explicit geometry dependence in the EM sector, then $F_{I, \text{extra}}^{(\text{EM})} = 0$.

(ii) Geometry force from plasma/matter loading. The matter-sector contribution depends on the matter model, but the geometry dependence is typically through a confinement/core potential $V_{\text{conf}}(\mathbf{X}; \Theta)$. Two common cases:

GNLS / field-theory matter: if the Hamiltonian contains $\sum_s \int \rho_s V_{\text{conf}}$, then

$$F_I^{(\text{mat})} = -\frac{\partial H_{\text{mat}}}{\partial \Theta^I} = -\sum_s \int d^3x dw \rho_s(t, \mathbf{x}, w) \partial_{\Theta^I} V_{\text{conf}}(\mathbf{x}, w; \Theta) + \dots \quad (277)$$

where $\dots$ denotes any additional explicit geometry dependence of EOS or dispersive terms (if present).

Vlasov / PIC matter: if particles experience a geometry-dependent potential V_{conf} , then in continuum form

$$H_{\text{mat}} \supset \sum_s \int f_s(t, \mathbf{x}, w, \mathbf{v}, v_w) V_{\text{conf}}(\mathbf{x}, w; \Theta) d^3x dw d^3v dv_w, \quad (278)$$

and hence

$$F_I^{(\text{mat})} = - \sum_s \int f_s \partial_{\Theta^I} V_{\text{conf}} d^3x dw d^3v dv_w = - \sum_s \int \rho_s(t, \mathbf{x}, w) \partial_{\Theta^I} V_{\text{conf}}(\mathbf{x}, w; \Theta) d^3x dw, \quad (279)$$

which is the same loading form, written either in phase space or using $\rho_s = \int f_s d^3v dv_w$.

(iii) **Ratio-penalty forces.** For (271), the explicit forces are

$$F_a^{(\text{ratio})} = - \frac{\partial E_{\text{ratio}}}{\partial a} = +\kappa\alpha (L - \alpha a), \quad F_L^{(\text{ratio})} = - \frac{\partial E_{\text{ratio}}}{\partial L} = -\kappa (L - \alpha a). \quad (280)$$

G.4 Dynamics: damped collective-coordinate evolution and quasi-static limits

A baseline (simulation-friendly) geometry evolution law is a damped second-order system:

$$M_{IJ} \ddot{\Theta}^J + \Gamma_{IJ} \dot{\Theta}^J = F_I, \quad (281)$$

with positive semidefinite damping matrix Γ_{IJ} . In diagonal form for $\Theta^I = (a, L)$, this becomes

$$M_a \ddot{a} + \Gamma_a \dot{a} = F_a, \quad M_L \ddot{L} + \Gamma_L \dot{L} = F_L. \quad (282)$$

Two controlled limits are frequently useful:

- **Overdamped relaxation:** $M_{IJ} \rightarrow 0$, yielding $\Gamma_{IJ} \dot{\Theta}^J = F_I$.
- **Adiabatic elimination / instantaneous minimization:** when geometry relaxation is much faster than plasma timescales, one sets

$$\frac{\partial H_{\text{tot}}}{\partial \Theta^I} \approx 0, \quad \text{i.e.} \quad F_I \approx 0, \quad (283)$$

and solves an algebraic equilibrium problem for $\Theta(t)$ given instantaneous plasma loads.

Optional: geometry fields on the brane. If one wants *spatially varying* localization/throat response, promote $\Theta^I(t) \rightarrow \Theta^I(t, \mathbf{x})$ and use a gradient-regularized relaxation PDE, e.g.

$$\tau_I \partial_t \Theta^I = - \frac{\delta H_{\text{tot}}}{\delta \Theta^I} + D_I \nabla_3^2 \Theta^I, \quad (284)$$

where $\delta/\delta\Theta^I$ is a functional derivative and the diffusion term enforces smoothness (prevents grid-scale localization-width noise). This is optional but can be valuable for modeling localized thinning/thickening that correlates with reconnection sites.

G.5 Worked example: dynamic Gaussian width $\lambda(t)$

Take

$$Z(w; \lambda) = \exp\left(-\frac{w^2}{\lambda^2}\right), \quad \partial_\lambda Z(w; \lambda) = \frac{2w^2}{\lambda^3} Z(w; \lambda). \quad (285)$$

Then the EM-driven generalized force from (276) becomes

$$F_\lambda^{(\text{EM})} = - \int d^3x dw \frac{1}{2\mu_0} \left(\frac{2w^2}{\lambda^3} Z(w; \lambda) \right) \left(E_A E_A + \frac{1}{2} F_{AB} F_{AB} \right). \quad (286)$$

This force is always nonpositive for nontrivial fields (it tends to reduce λ) unless balanced by $E_{\text{geom}}(\lambda)$ or matter contributions that favor expansion. A stable $\lambda_\star$ requires competing effects in H_{tot} .

Impact on reduced brane parameters. When λ changes, both the effective Maxwell coupling and KK masses change:

$$\mu_0^{\text{eff}}(\lambda) = \frac{\mu_0}{\int dw Z(w; \lambda)} = \frac{\mu_0}{\lambda\sqrt{\pi}}, \quad m_n^2(\lambda) = \frac{2n}{\lambda^2}. \quad (287)$$

In a $3D \times \text{modes}$ code, this means the diagonal mode operator $(\square_4 + m_n^2)$ acquires a time-dependent mass term. If $\lambda(t)$ varies slowly relative to the EM wave timestep, this is straightforward to implement as a time-dependent coefficient.

G.6 Worked example: tube DOFs $(a(t), L(t))$ and a minimal geometry energy

To model breathing/ringing of a localized tube/throat, one may use $\Theta = (a, L)$ with a minimal geometry energy (volume work plus surface tension):

$$E_{\text{geom}}(a, L) = P_{\text{vac}} V(a, L) + \sigma A(a, L), \quad (288)$$

where a natural 4D tube measure for a corridor $B^3(a) \times [0, L] \subset \mathbb{R}^4$ is

$$V(a, L) = \frac{4\pi}{3} a^3 L, \quad A(a, L) = 4\pi a^2 L + 2 \cdot \frac{4\pi}{3} a^3. \quad (289)$$

The resulting purely geometric restoring forces are

$$F_a^{(\text{geom})} = -\frac{\partial E_{\text{geom}}}{\partial a}, \quad F_L^{(\text{geom})} = -\frac{\partial E_{\text{geom}}}{\partial L}, \quad (290)$$

with explicit derivatives obtained directly from (288)–(289).

The plasma loads enter through $F_a^{(\text{mat})}$ and $F_L^{(\text{mat})}$ via $\partial_{a,L} V_{\text{conf}}$, and (optionally) through $F_{a,L}^{(\text{EM})}$ if the EM sector is given explicit geometry dependence (e.g. geometry-dependent Z , geometry-dependent imposed drive).

G.7 How this is used in massive plasma simulations

In large runs, geometry DOFs are intended to be cheap and ledger-consistent. A practical operator-split update per timestep Δt is:

1. **Advance plasma and EM fields** from $t^n \rightarrow t^{n+1}$ using the current geometry Θ^n (i.e. current $Z(w; \Theta^n)$, $V_{\text{conf}}(\cdot; \Theta^n)$, and mode masses if applicable).
2. **Compute instantaneous geometry forces** F_I^n using (273) with numerical quadrature in w . In a $3D \times \text{modes}$ solver, the EM force contribution can be computed from mode energies because H_{EM} is a sum of diagonal mode contributions plus any explicit Θ -dependence.
3. **Update geometry** $\Theta^n \rightarrow \Theta^{n+1}$ using (281) with a stable ODE integrator (explicit with damping is often sufficient; implicit is optional if stiffness is large).
4. **Update derived coefficients** (e.g. $\mu_0^{\text{eff}}(\Theta)$, $m_n(\Theta)$, coupling weights $c_n(\Theta)$) for the next field step.

Energy accounting. If $\Gamma_{IJ} = 0$ and the time integration is done conservatively, the combined field+geometry energy (272) is conserved up to numerical error. With $\Gamma_{IJ} \neq 0$, geometry damping removes energy from the resolved system; this can be interpreted as coupling to unresolved internal wall modes and should be tracked explicitly in the simulation energy ledger.

Why this matters for beyond-MHD physics. Allowing Θ (and hence Z , confinement, and transverse spectra) to respond to plasma loads provides a physically grounded mechanism for apparent non-ideality on the brane: magnetic energy and helicity can be exported into transverse and geometry reservoirs (Appendix E) rather than being represented solely by an *ad hoc* resistivity. This is precisely the kind of channel needed to model reconnection-like phenomena while keeping the bulk dynamics conservative and diagnosable.